

Digital Communication System



Unit 3: Cellular Communication System

3.1 Mobility, Mobile and Ubiquitous computing

3.2 Global System for Mobile Communication (GSM) system overview:

3.2.1 Cellular concept

3.2.2 GSM Architecture

3.2.3 Frequency Reuse Planning and Design

3.2.4 Mobility Management(Hard Handoff)

3.3 General Packet Radio Service (GPRS) architecture and working

3.4 Wireless Local Loop (WLL)

3.5 Introduction of 3G technology

3.5.1 Introduction of CDMA

3.5.2 Frequency Allocation

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Mobile Computing vs. Ubiquitous Computing

Mobile computing is when you bring the computer with you. Example smartphones, prisoner anklets.

Ubiquitous computing is wherever you go, there is a computer you can use. Example internet cafe, smart devices.



Mobile Computing

- **Mobile means** movable
- **Computing means** Computer

Definition : Mobile computing refers to the set of IT technologies, products, services and operational strategies and procedures that enable end users to access computation, information and related resources and capabilities while mobile(**in motion**).



Key Benefits of Mobile Computing

- The key advantage of mobile computing is convenience, where users have access to information and computational resources anytime and anywhere.
- The user is not restricted to a given geographic location.



Mobile Computing

It is a combination of three technologies, namely:

- **Mobile Hardware:**

This technology gives small powerful device and display with low energy consumption.

- **Mobile Software:**

Used to create Mobile Applications.

- **Wireless Communication Network:**

This technology provides higher bandwidth, higher data transfer rate at lower costs and with world wide roaming.

Applications of Mobile Computing

- Access to the web or the internet.
- GPS (Global Positioning System).
- Emergency services (hire an ambulance, police call, etc)
- Services relating to entertainment.
- Educational programs.
- Social media live interactions.

Characteristics of Mobile Computing

- **Portability** – The ability to easily switch a computer inside a learning environment or between environments.
- **Social Interactivity** – The opportunity for users to exchange data and collaborate with one another.
- **Context Sensitivity** – The ability to collect and respond to data that is specific to a particular place, climate, or time.
- **Connectivity** – The capacity to be remotely linked in any location for the purpose of data communication.

Characteristics of Mobile Computing

- **Individual** – The opportunity to use technology to provide platform and lesson customization for individual learners on challenging tasks.
- **Small Size** – Because of their phone-like dimensions, mobile devices are also known as handhelds, palmtops, and smartphones. A standard mobile device would fit in the hand or pocket of an average adult.
- **Wireless Connectivity** – Mobile devices may usually communicate with other mobile devices, as well as stationary computers and systems, networks, and mobile phones.

Advantages of Mobile Computing

- **Seamless and reliable communication**

- **Increased Productivity**- Mobile devices can be used in the field by a variety of businesses, saving time and money for both clients and employees.

- **Entertainment**- Mobile devices may be used for entertainment, personal use, and even presentations to clients and colleagues.

Advantages of Mobile Computing

- **Portability**- one of the key benefits of mobile computing is that you are not limited to a single location in order to complete tasks or access email when on the go.
- **Cloud Computing**- This program allows you to save data on an online server and access them from any computer with an internet connection. You can also access these files on several mobile devices.

Disadvantages of Mobile Computing

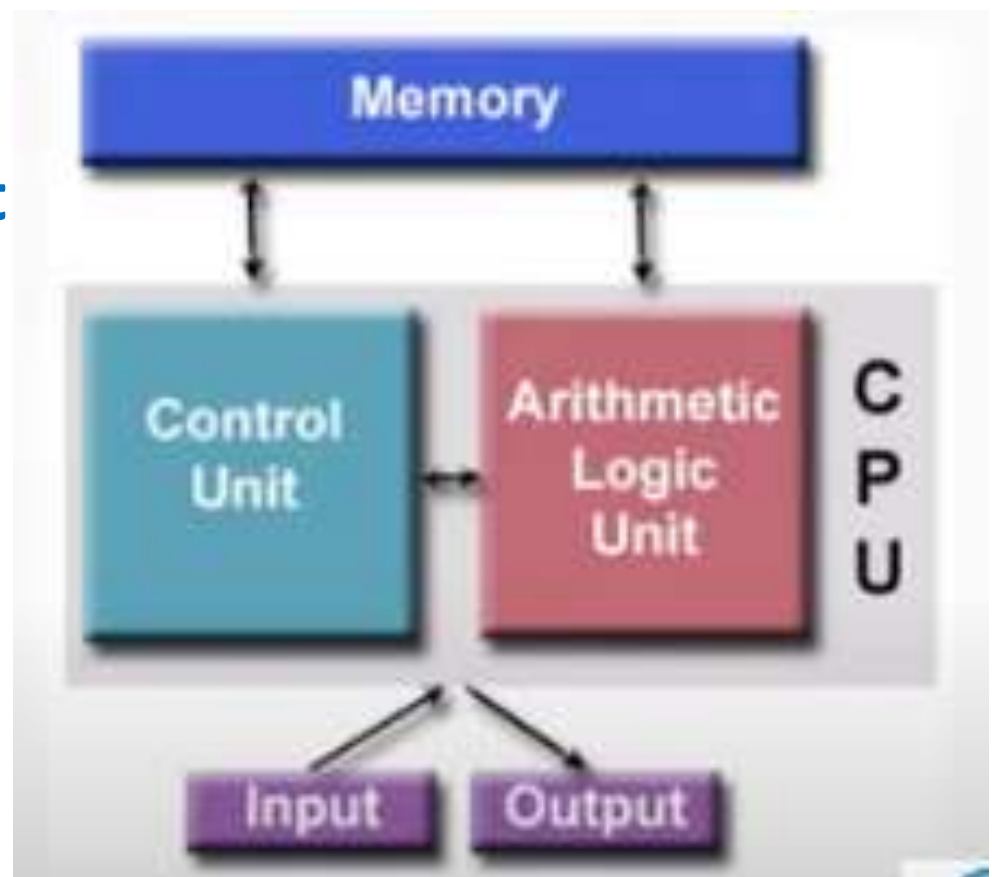
- The obstacle to battery consumption.
- The transmission bandwidth is inefficient.
- Over the whole network, there are link losses.
- Fluctuation in the stability of the network.
- Small screen sizes.
- The issue of interoperability.

Ubiquitous Computing

- **What is a computer?**

- A computer is a device which takes input, process data and produces output.

- **Sense -> Process -> Result**



Ubiquitous Computing

- **Ubiquitous means** Present Everywhere
- **Computing means** Computer
- **Definition** : It is also called as **Pervasive computing**, and it is the new trend toward embedding everyday objects with microprocessors so that they can communicate information.



Ubiquitous Computing

- It refers to the **presence of computers in common objects found all around us** so that people are unaware of their presence. All these devices communicate with each other over wireless networks without the interaction of the user.



Working of Ubiquitous Computing

- To implement this computing concept we need smart objects. We can convert any ordinary object around us into a smart object (like fan, bulb, motor, AC, etc). We just need to integrate 3 properties into these objects.

- Sense**

- use sensors

- Process**

- use electronic circuits

- Result**

- may be in electronic form or mechanical form



Ubiquitous Computing

Pervasive computing is a combination of three technologies, namely:

- **Micro electronic technology:**

This technology gives small powerful device and display with low energy consumption.

- **Digital communication technology:**

This technology provides higher bandwidth, higher data transfer rate at lower costs and with world wide roaming.

- **The Internet standardization:**

This standardization is done through various standardization bodies and industry to give the framework for combining all components into an interoperable system with security, service and billing systems.

Applications Ubiquitous Computing

Personal Identification and Authentication

Health Care

Transport System

Home Automation

Defense Industry

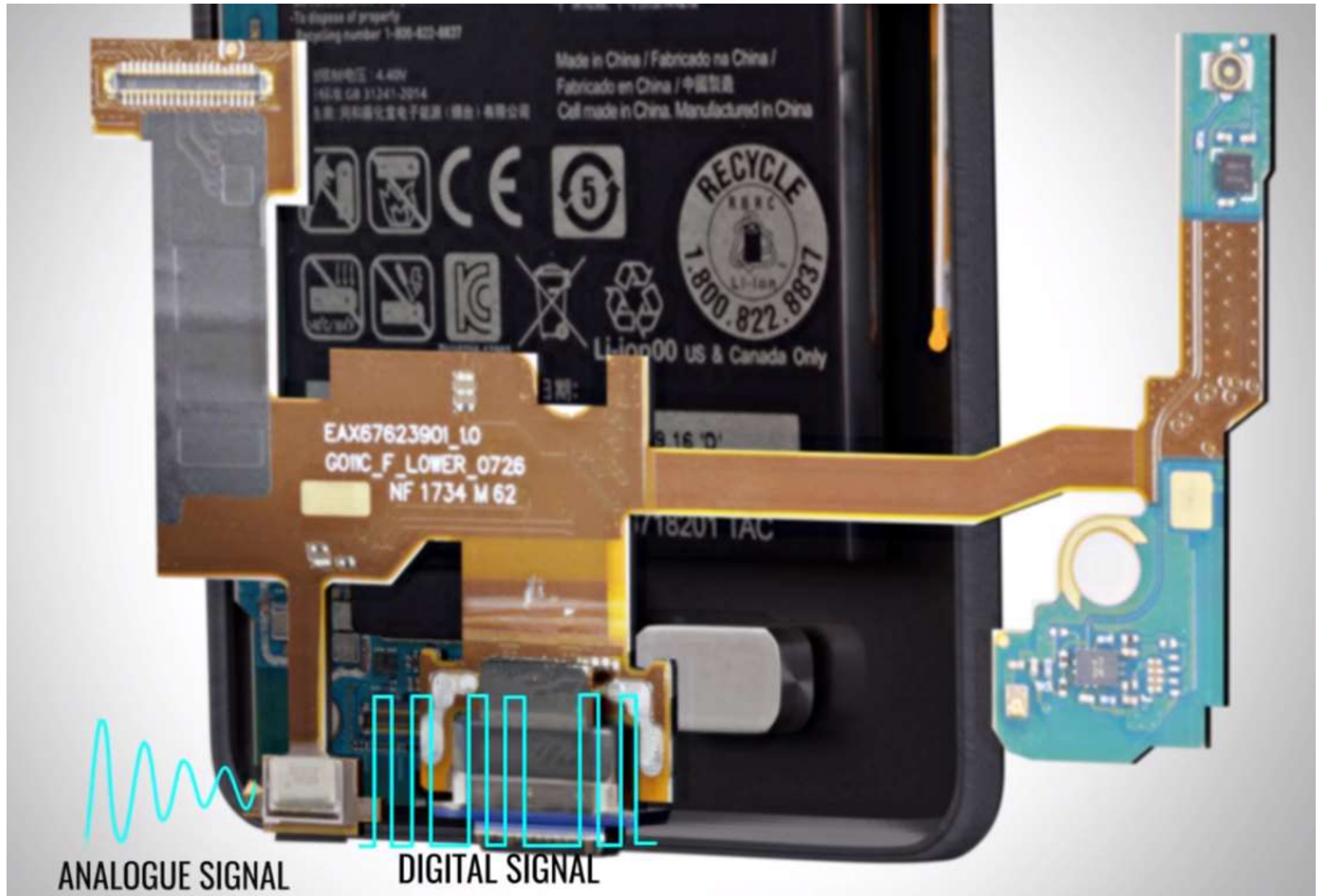
Industrial Production and Material management

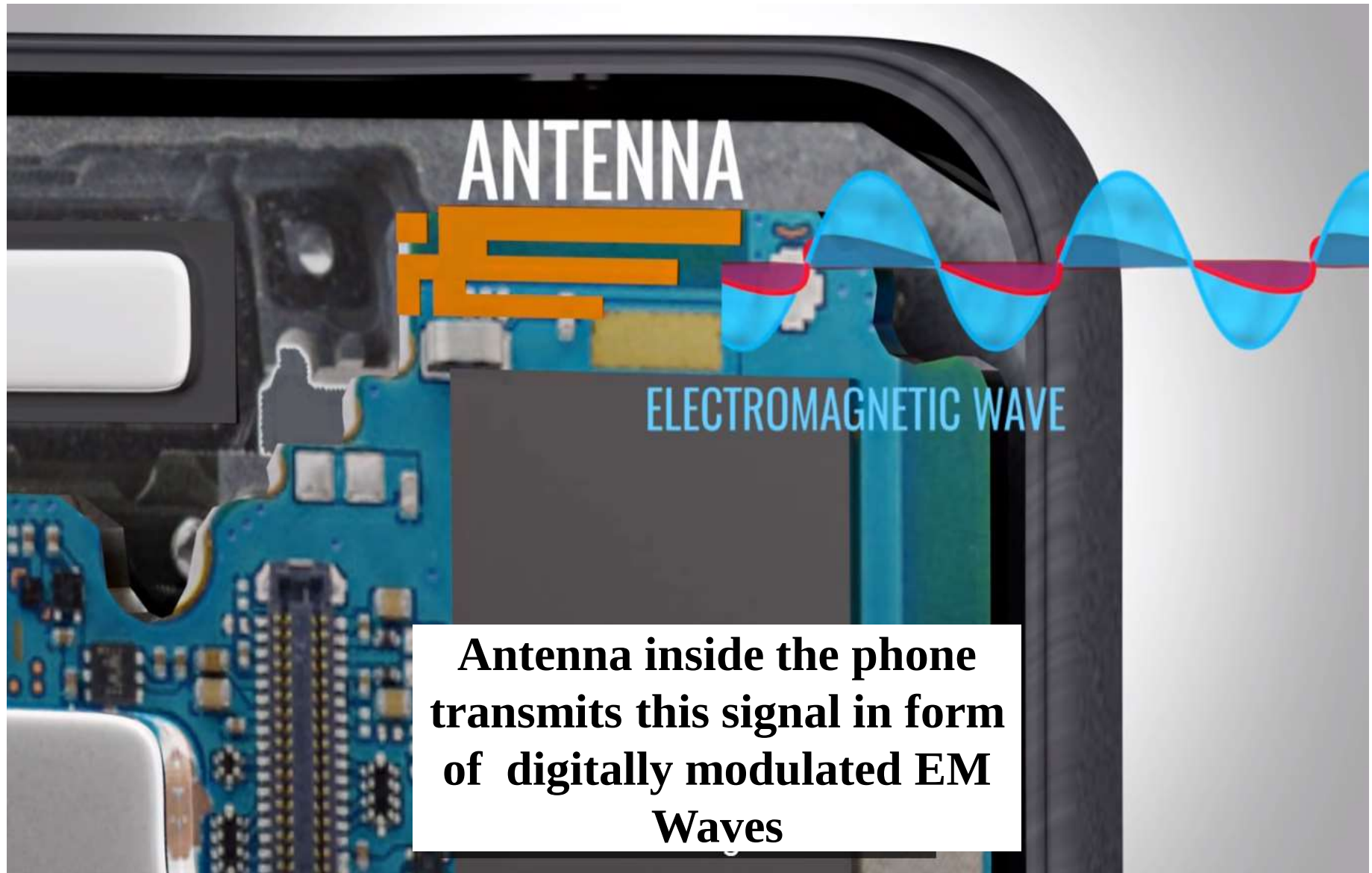
Global System for Mobile Communication (GSM)



**The microphone converts
your voice into the digital
signal**







ANTENNA

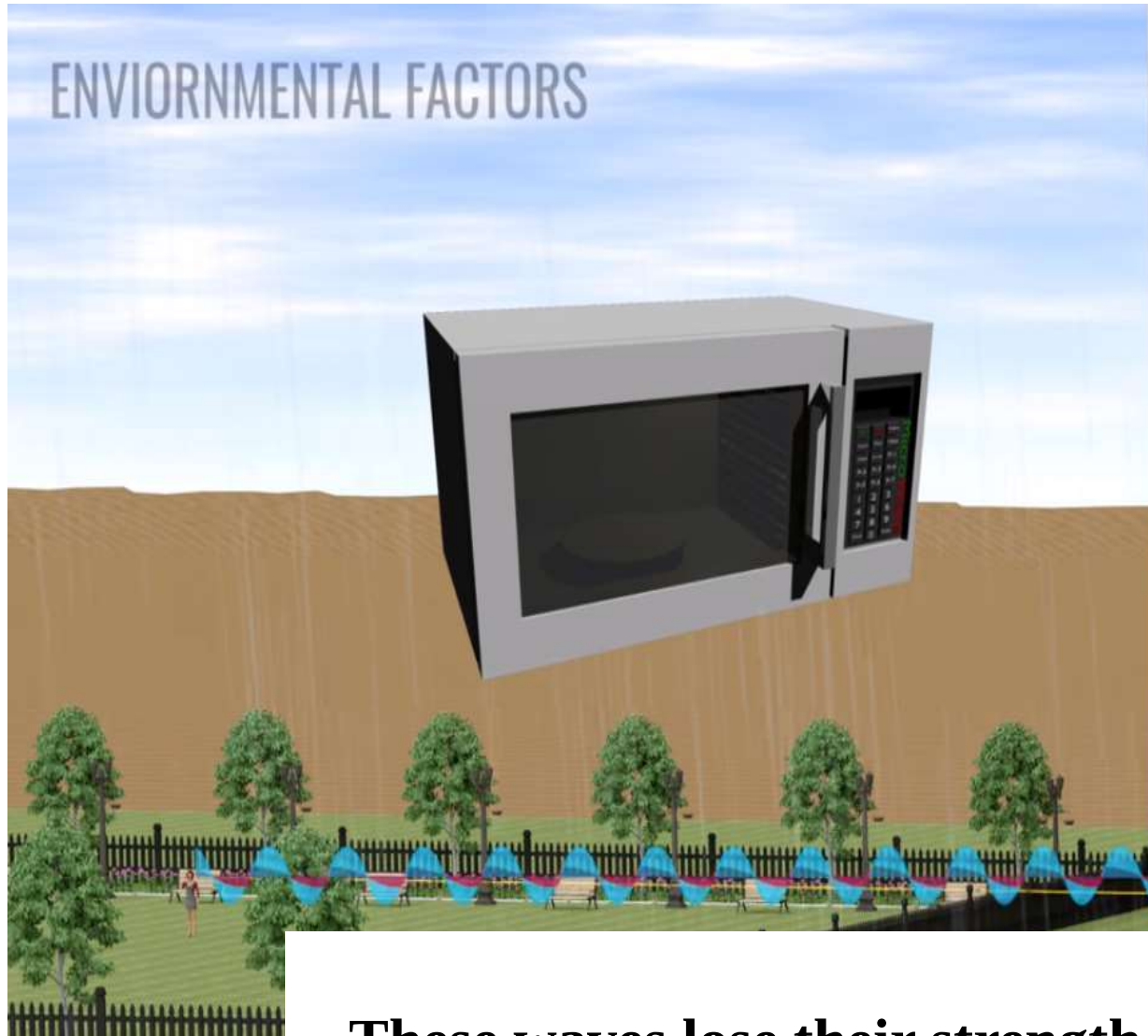
ELECTROMAGNETIC WAVE

**Antenna inside the phone
transmits this signal in form
of digitally modulated EM
Waves**

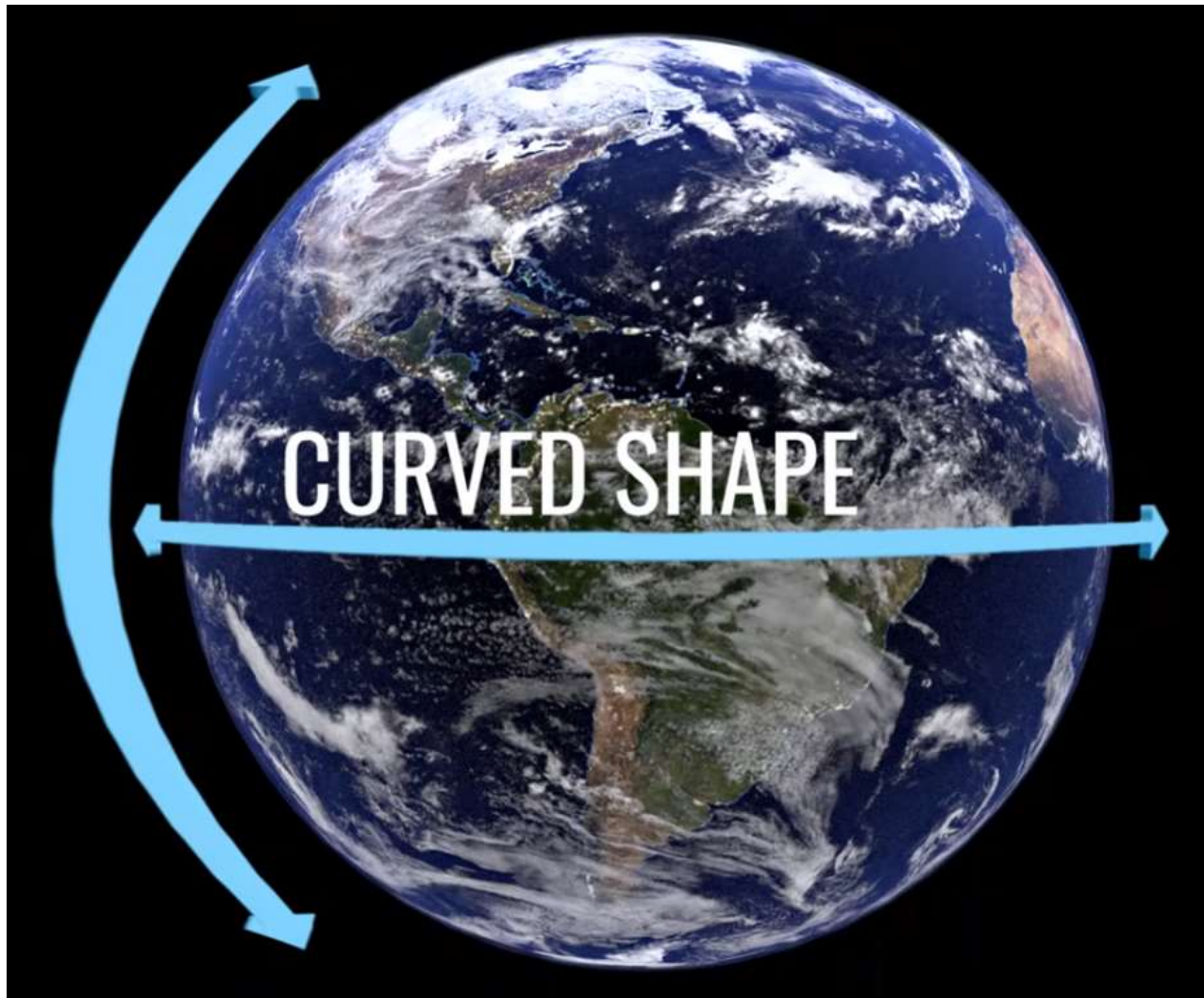


by using low and high
frequencies respectively.

ENVIRONMENTAL FACTORS



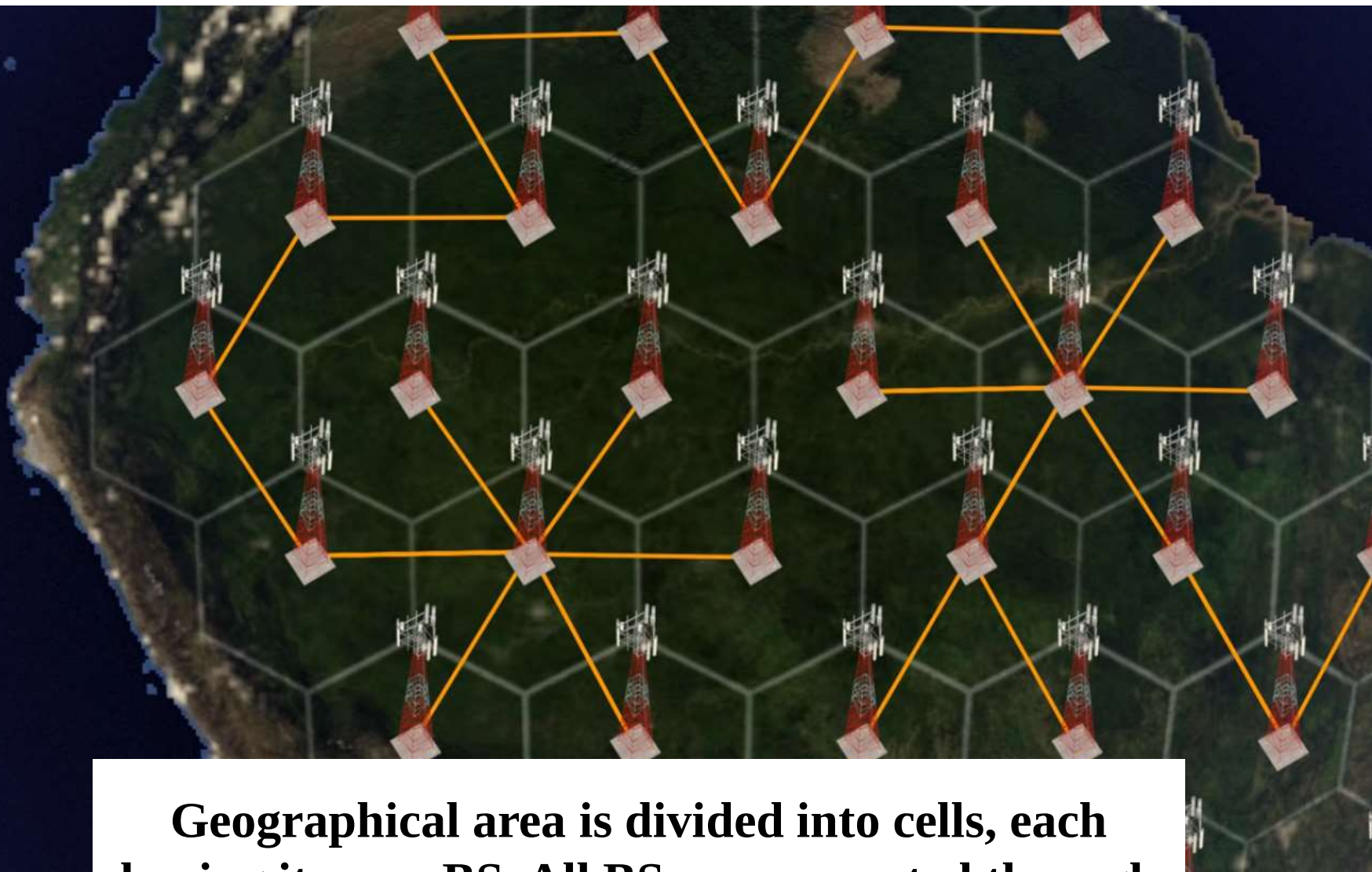
These waves lose their strength due to physical objects and climatic conditions



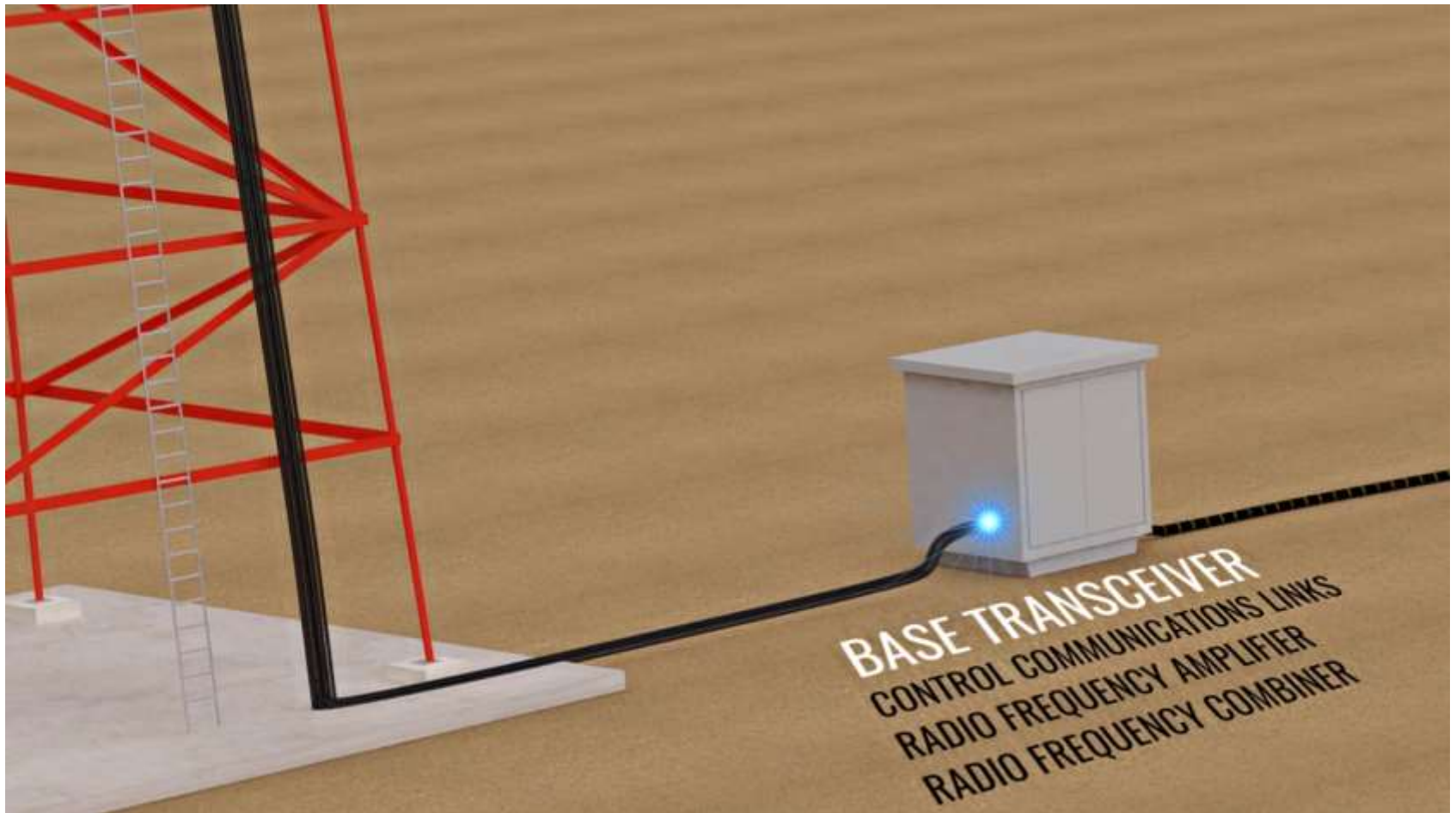
**They are also affected by
Earth's curvature**



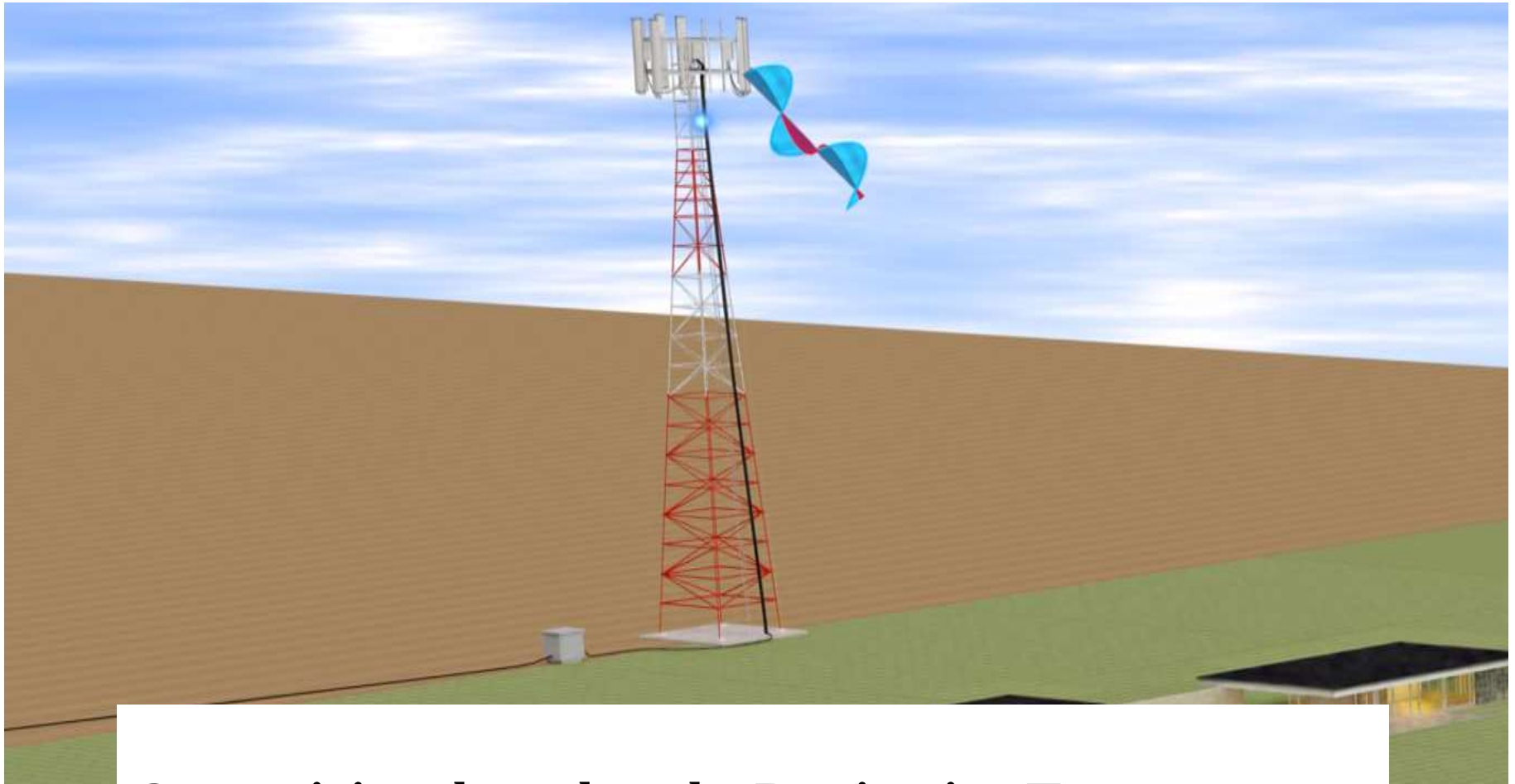
**To overcome these issues,
Cell Towers were
introduced**



Geographical area is divided into cells, each having its own BS. All BS are connected through Optic fibre cable underground or under ocean.



EM Waves received by the antenna are converted to Light pulses which pass through the Base Transceiver. Here it routes your call to Destination Tower



On receiving the pulses the Destination Tower converts it into EM Waves and the receiver receives your call.

HOW TO LOCATE YOUR FRIEND'S TOWER ?



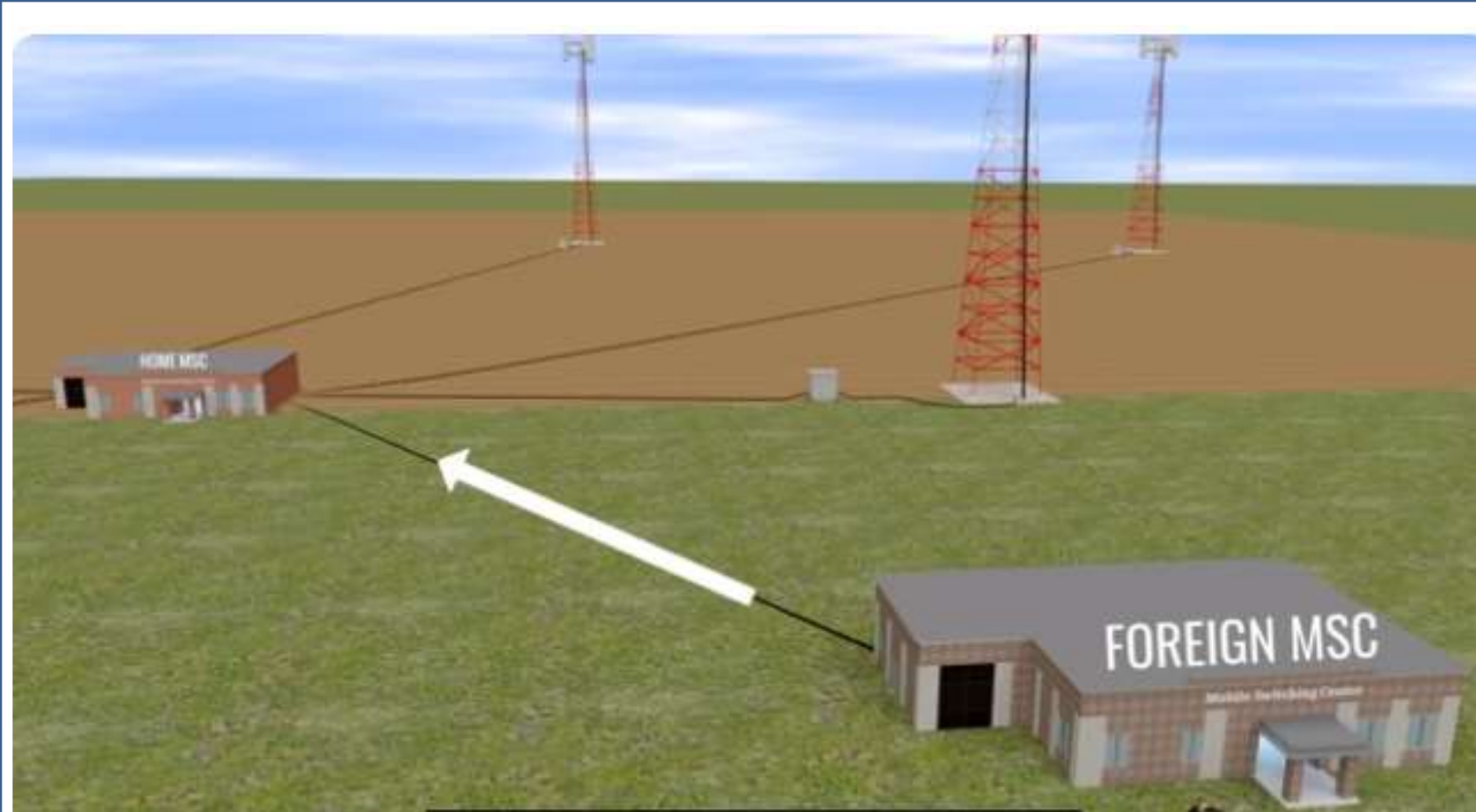
MOBILE SWITCHING CENTER (MSC)



The MSC is the central point of a group of cell towers.



All your subscription information is stored in a SIM card in your mobile phone which in turn stores all the details in the MSC. This MSC is your Home MSC. Home MSC stores details like subscription plan, activity status, your current location,



Your Home MSC always knows your foreign MSC where you are currently located. It is done by periodically updating the location information of your phone, when your phone is turned on or when you cross towers the location update is done.

Frequency Spectrum



Frequency range is quite limited for cellular communication and there are billions of subscribers. This issue is solved with the help of two technologies.

1. FREQUENCY SLOT DISTRIBUTION

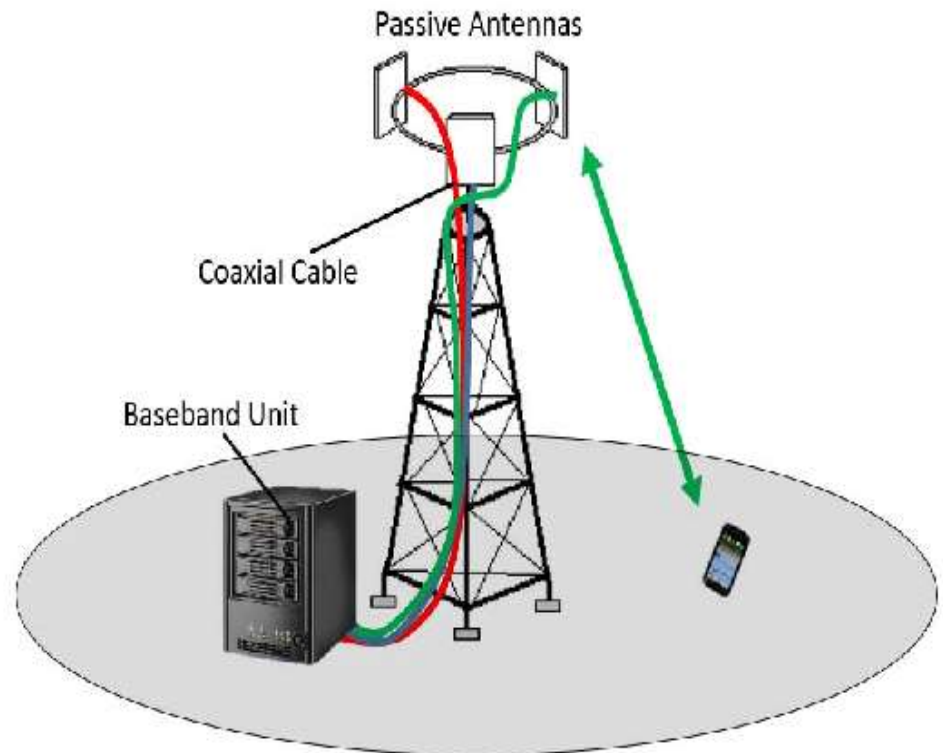
2. MULTIPLE ACCESS TECHNIQUE

Frequency Slot Distribution technique is used to carefully allocate different frequency slots to each cell tower in the network. In Multiple access techniques these frequencies are efficiently distributed among the users in that cell tower.

Cellular Network

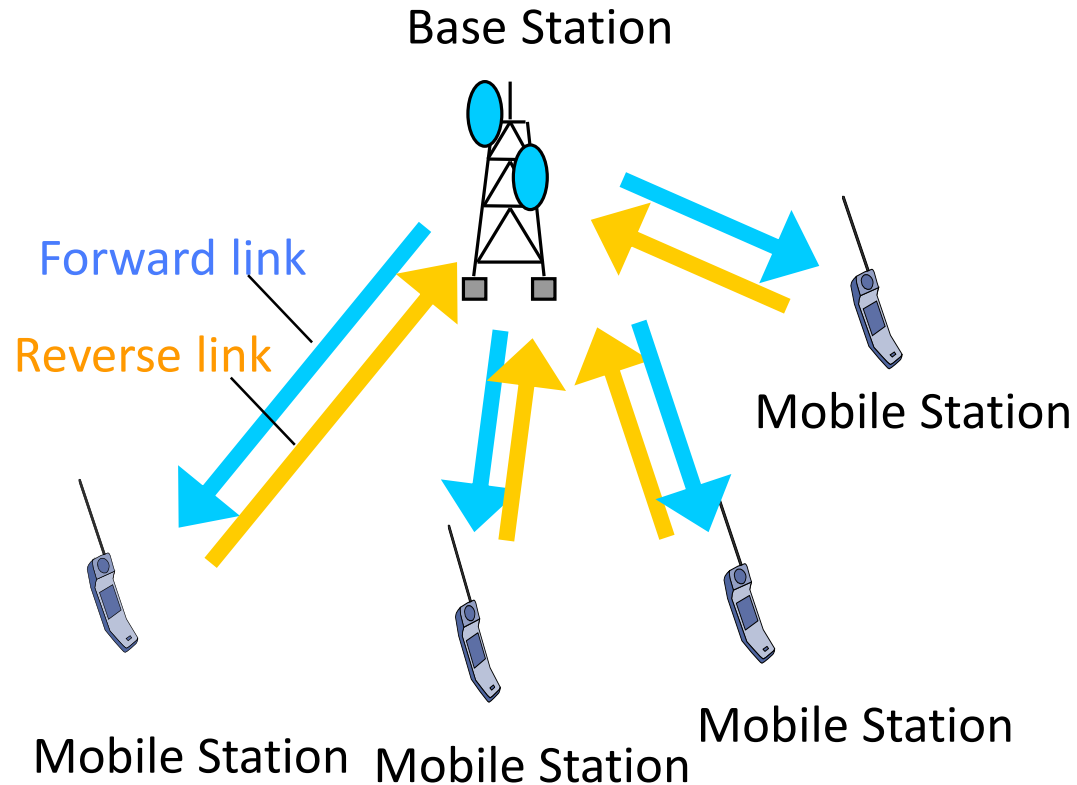
A Cellular network or Mobile network **is a radio network distributed over land areas called cells**, each served by at least one fixed-location transceiver, known as a cell site or **base station**.

Cellular networks use lower power (100w), shorter range and more transmitters for data transmission.



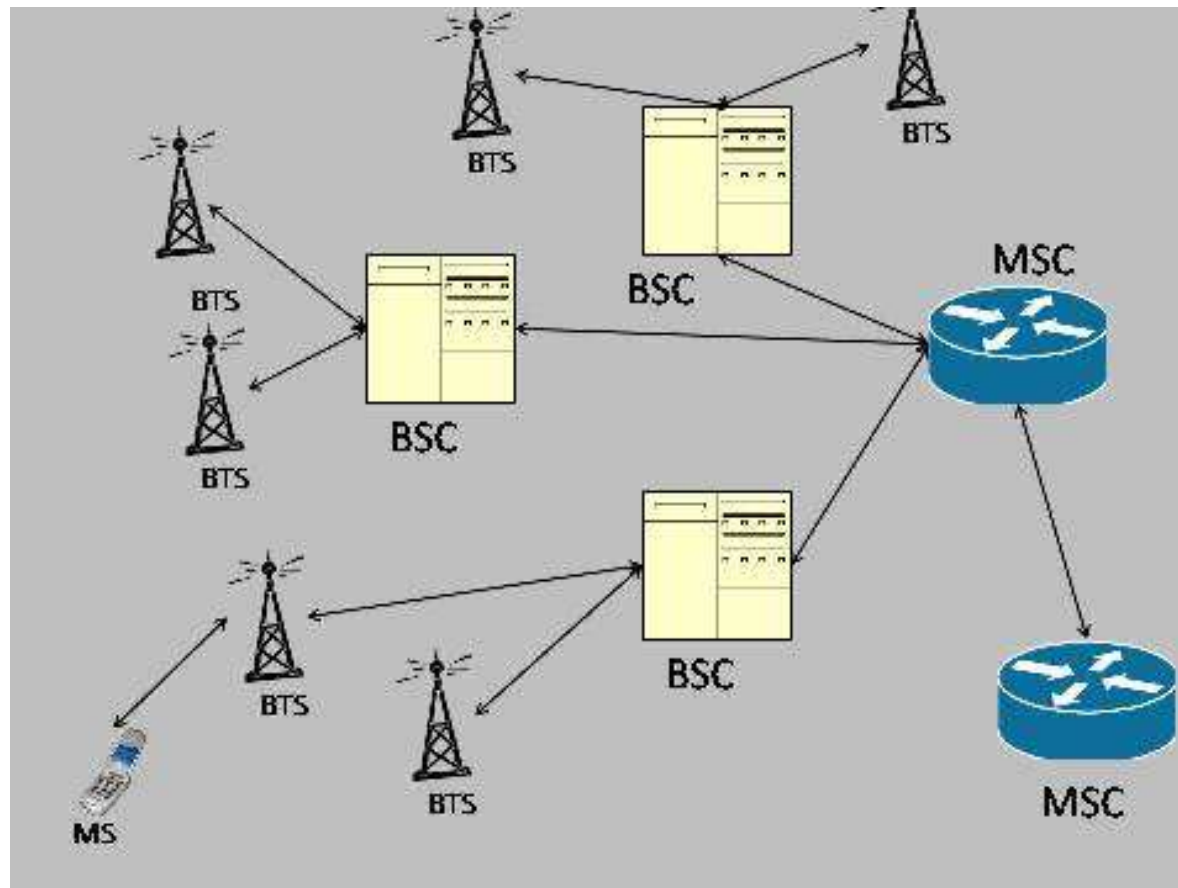
Cellular Network

- The base station communicates with mobiles via a channel
- The channel is made of two frequencies, one for transmitting to the base station and one to receive information from the base station

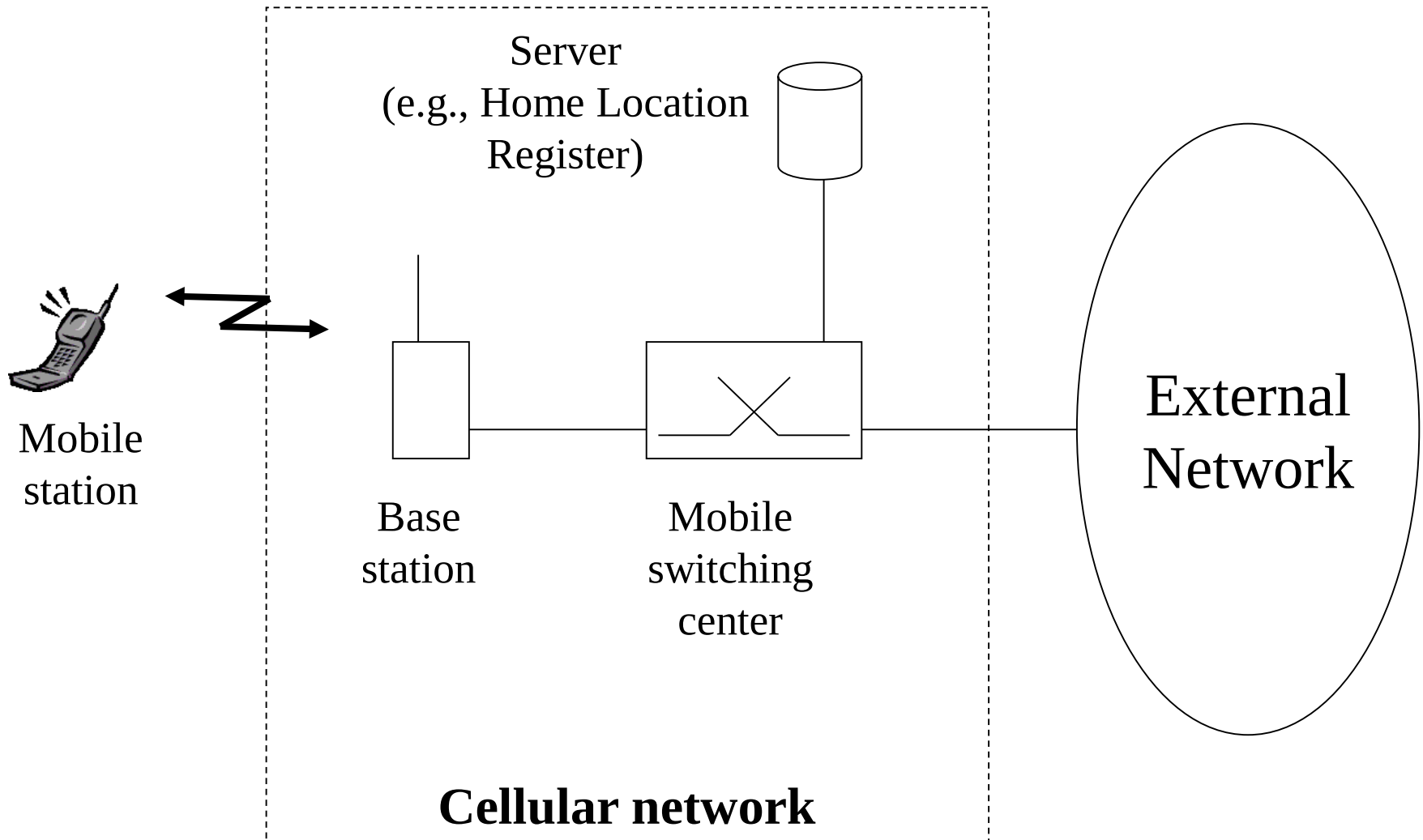


Cellular Network

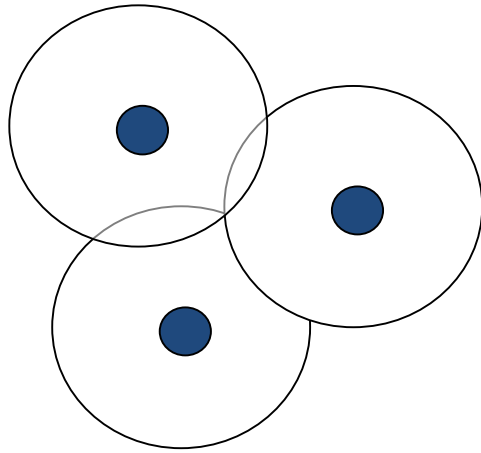
All these base stations are connected to **Mobile Switching Center (MSC)** which connects cells to wide area net, manages call setup and handles mobility.



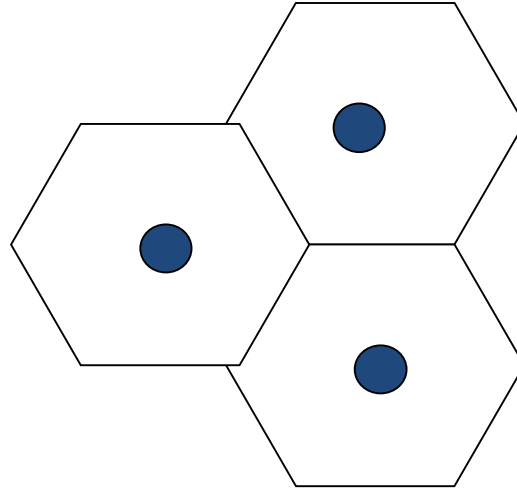
Basic Architecture of Cellular Network



Representation of Cells



Ideal cells



Fictitious cells

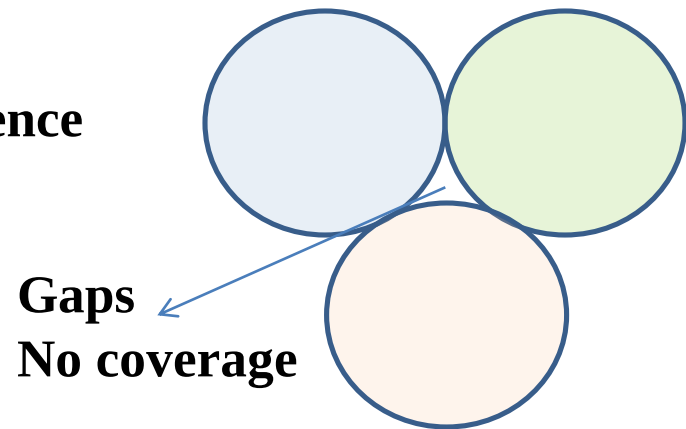
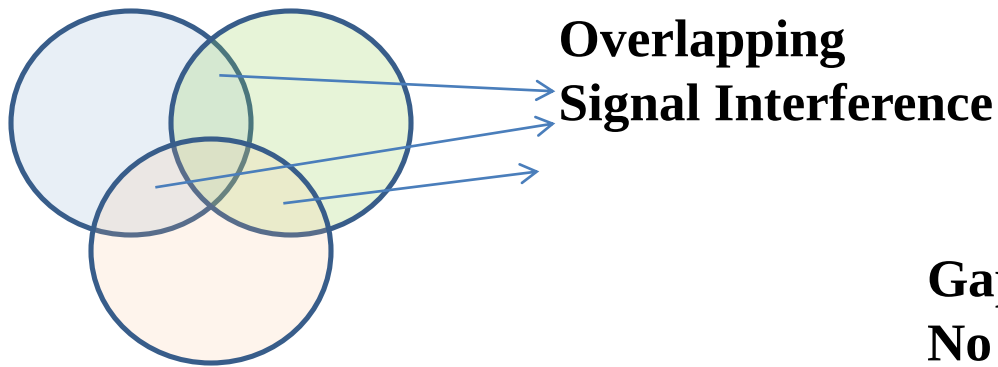
Radio signal transmission in reality is **circular shape**.
But For calculation purpose we will use **hexagonal shape**.

Why Cell Shape is Hexagonal?

In Cellular System land area is divided into regular shaped cells. Maximum area is covered by a circle but still we don't consider this shape because of two reasons:

Gaps between circles -> no coverage

Overlapping of circles -> Signal interference

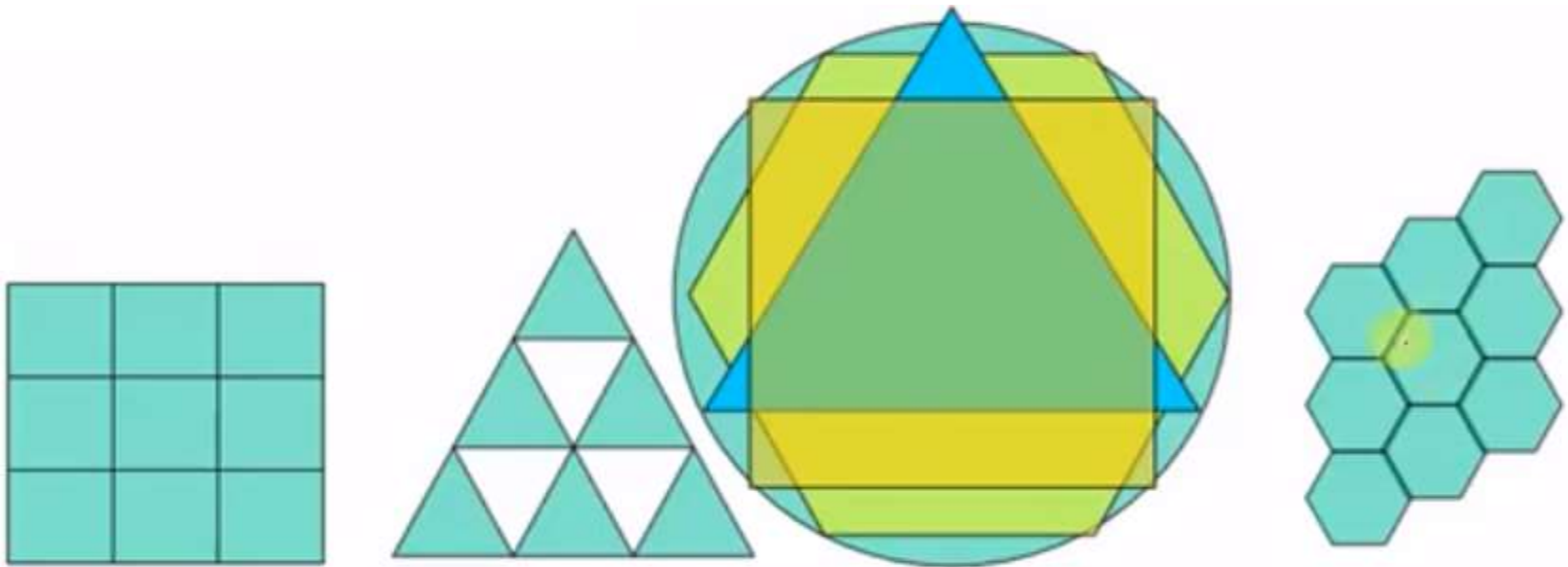


Why Cell Shape is Hexagonal?

In Cellular System land area is divided into regular shaped cells.

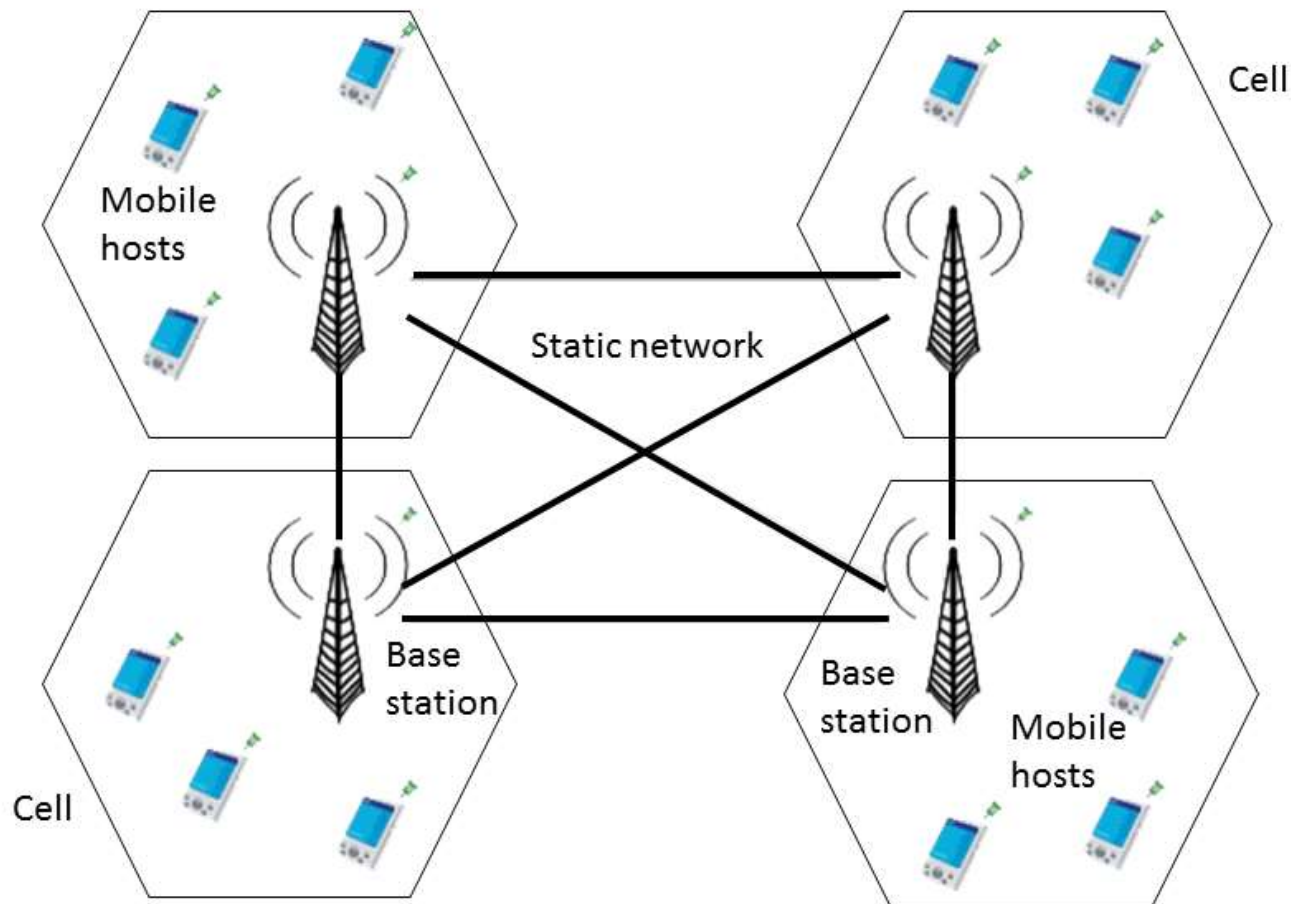
All the below criteria should be fulfilled for cell shape :

- 1. It should be a geometric shape (Circle, Square, Triangle, Hexagon)**
- 2. Area without overlap (Square, Triangle, Hexagon)**
- 3. It should cover maximum area (Area of Hexagon is highest)**



Why Cell Shape is Hexagonal?

Hexagon is the only shape which fulfills all the criteria well, therefore cell shape is Hexagonal.



Need of different size of Cells

- Extending the coverage to the areas that are difficult to cover by a large cell.
- Increasing the capacity of the network for those areas that have a higher density of users.
- Increasing number of wireless devices and the communication between them.

Cell Hierarchy

Femtocells: Smallest unit of the hierarchy, these cells need to cover only a few meters where all devices are in the physical range of the users.

Picocells: Size of these networks is in the range of few tens of meters, e.g., WLANs.

Microcells: Cover a range of hundreds of meters e.g. in urban areas to support PCS which is another kind of mobile technology.

Cell Hierarchy

Macro cells: Cover areas in the order of several kilometers, e.g., cover metropolitan areas.

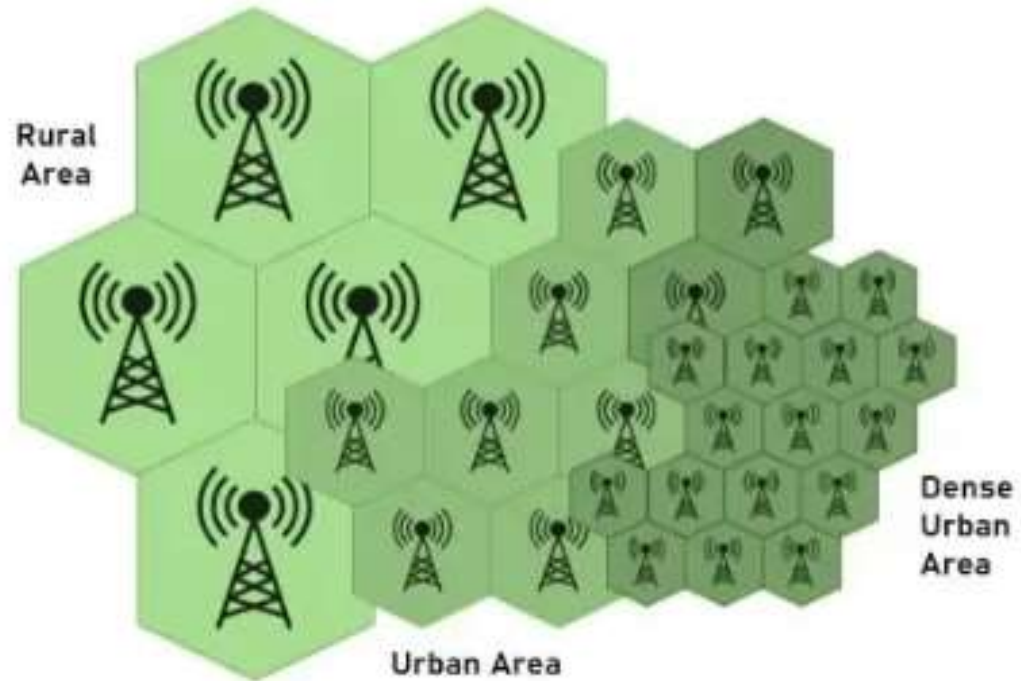
Mega cells: Cover nationwide areas with ranges of hundreds of kilometers, e.g., used with satellites.

Cell Hierarchy

The size of the cell varies based on how many cells are made within that geographical area.

The smallest cell has the highest population density.

Largest cells are located in rural areas.



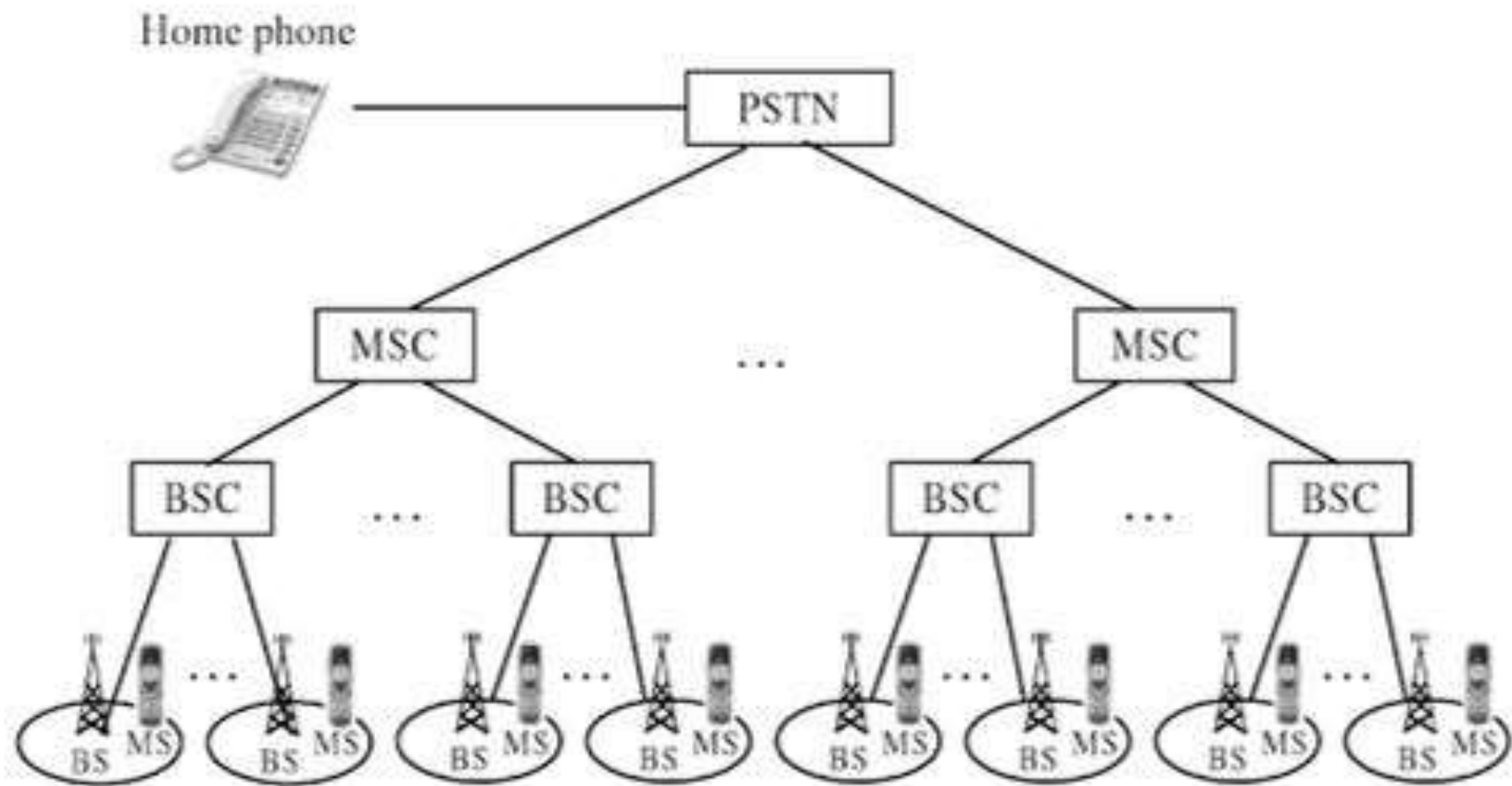


Fig: cellular system infrastructure

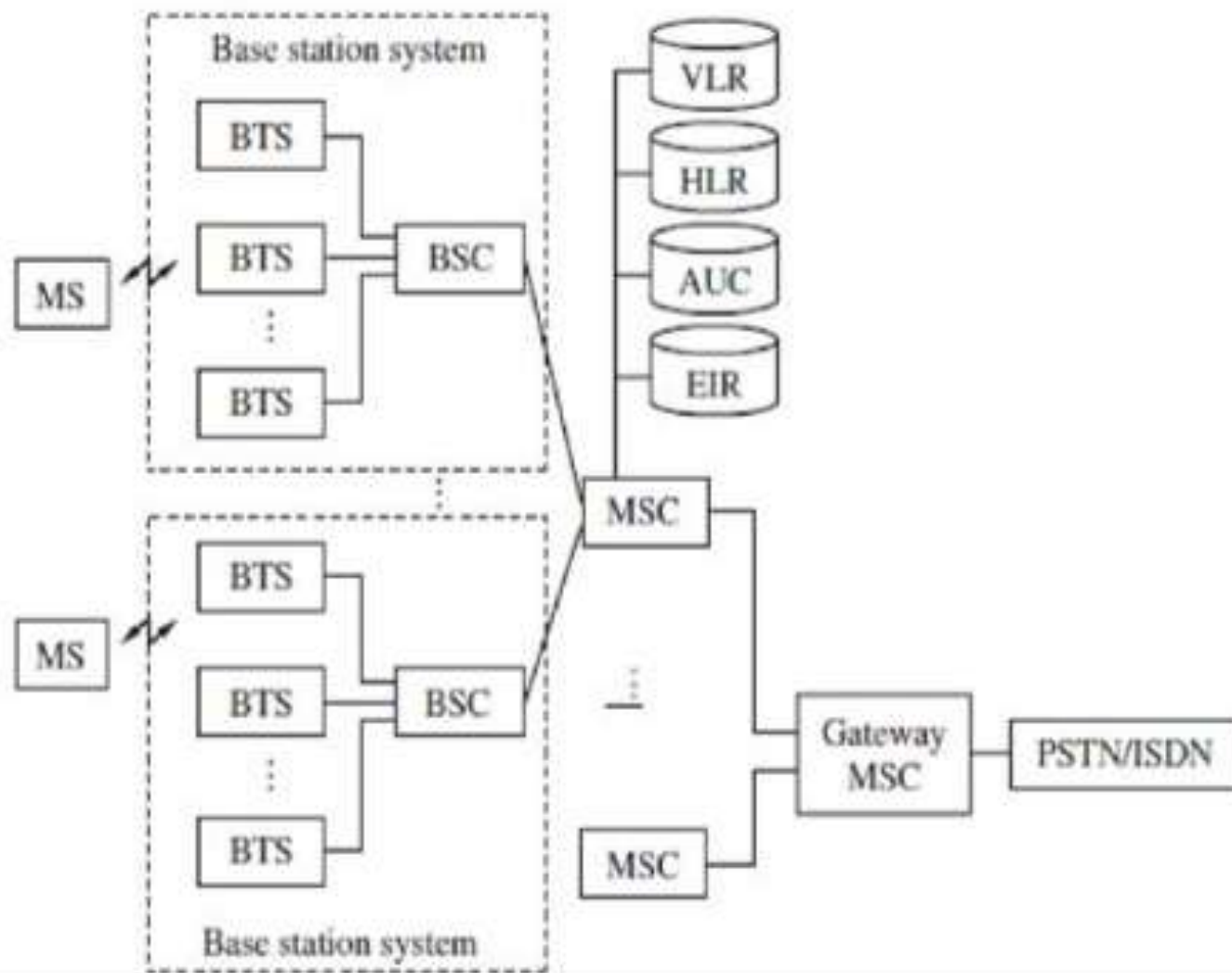
Cellular Network Architecture

- In a cellular structure, a MS (mobile station) needs to communicate with the BS of the cell where the MS is currently located and the BS acts as a gateway to the rest of the world.
- Therefore, to provide a link, the MS needs to be in the area of one of the cells (BS) so that mobility of the MS can be supported.

Cellular Network Architecture

- Several base stations are connected through hard-wires and are controlled by a BS controller (BSC), which in turn is connected to a mobile switching center (MSC).
- Several mobile switching centers are interconnected to a PSTN (public switched telephone network) and the ATM (asynchronous transfer mode) backbone.

Cellular Network Architecture



MS - Mobile Station

MSC - Mobile Switching Center

BSC - Base Station Controller

BTS - Base Transceiver System

VLR - Visitor Location Register

AUC - Authentication Center

HLR - Home Location Register

EIR - Equipment Identity Register

PSTN - Public Switching Telephone Network

Cellular Network Architecture

Mobile Station (MS)

- MS consists of following two components
 - **Mobile Equipment (ME)** – Handset you use
 - **Mobile Subscriber Identity Module (SIM)** - The SIM provides personal mobility so that the user can have access to all subscribed services irrespective of both the location of the terminal and the use of a specific terminal.

Cellular Network Architecture

Base Transceiver System (BTS)

- The BTS corresponds to the transceivers and antennas used in each cell of the network. A BTS is usually placed in the center of a cell and handles the radio link protocols with the MS.
- Functions of BTS are **encoding, encrypting, multiplexing, modulating, and feeding the RF signals to the antenna.**

Cellular Network Architecture

Base Station Controller (BSC)

- One BSC can handle multiple BTS.
- Functions of BSC are :
 - to handle radio channel setup, frequency hopping, and handovers.
 - The function of the BSC is to allocate the necessary time slots between the BTS and the MSC.

Cellular Network Architecture

Mobile Switching Centre (MSC)

- It is a heart of GSM network. Every MSC is identified by a unique ID.
- The MSC performs the switching of calls between the mobile and other fixed or mobile network users, as well as the management of mobile services such as registration, authentication, location updating, handovers, and call routing to a roaming subscriber, call recording, call tracing, etc.

Cellular Network Architecture

Equipment Identity Register (EIR)

The Equipment Identity Register (EIR) is a database that contains a list of all valid mobile equipment on the network, where its **International Mobile Equipment Identity (IMEI)** identifies each MS. An IMEI is marked as invalid if it has been reported stolen or is not type approved.

Cellular Network Architecture

Authentication Register (AUC)

- The AUC (authentication center) unit provides authentication and encryption parameters that verify the user's identity and ensure the confidentiality of each cell.
- The AUC protects network operators from different types of fraud found in today's cellular world.

Cellular Network Architecture

Home Location Register (HLR)

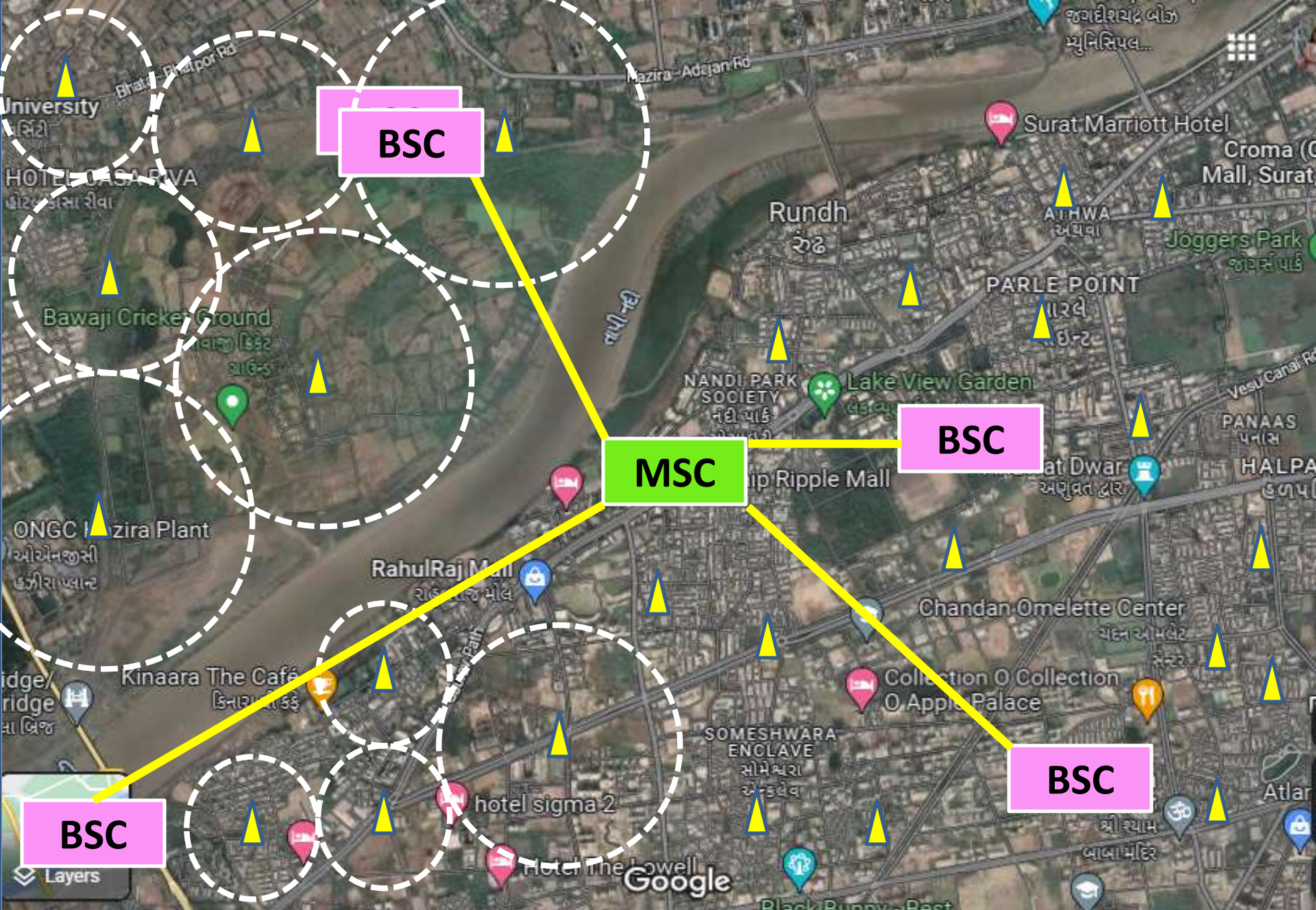
- The HLR is a database used for storage and management of subscriptions. The HLR is considered the most important database, as it stores permanent data about subscribers, including a subscription plan, his caller tune, location information, and activity status.
- When an individual buys a subscription in the form of SIM, then all the information about this subscription is registered in the HLR of that operator.

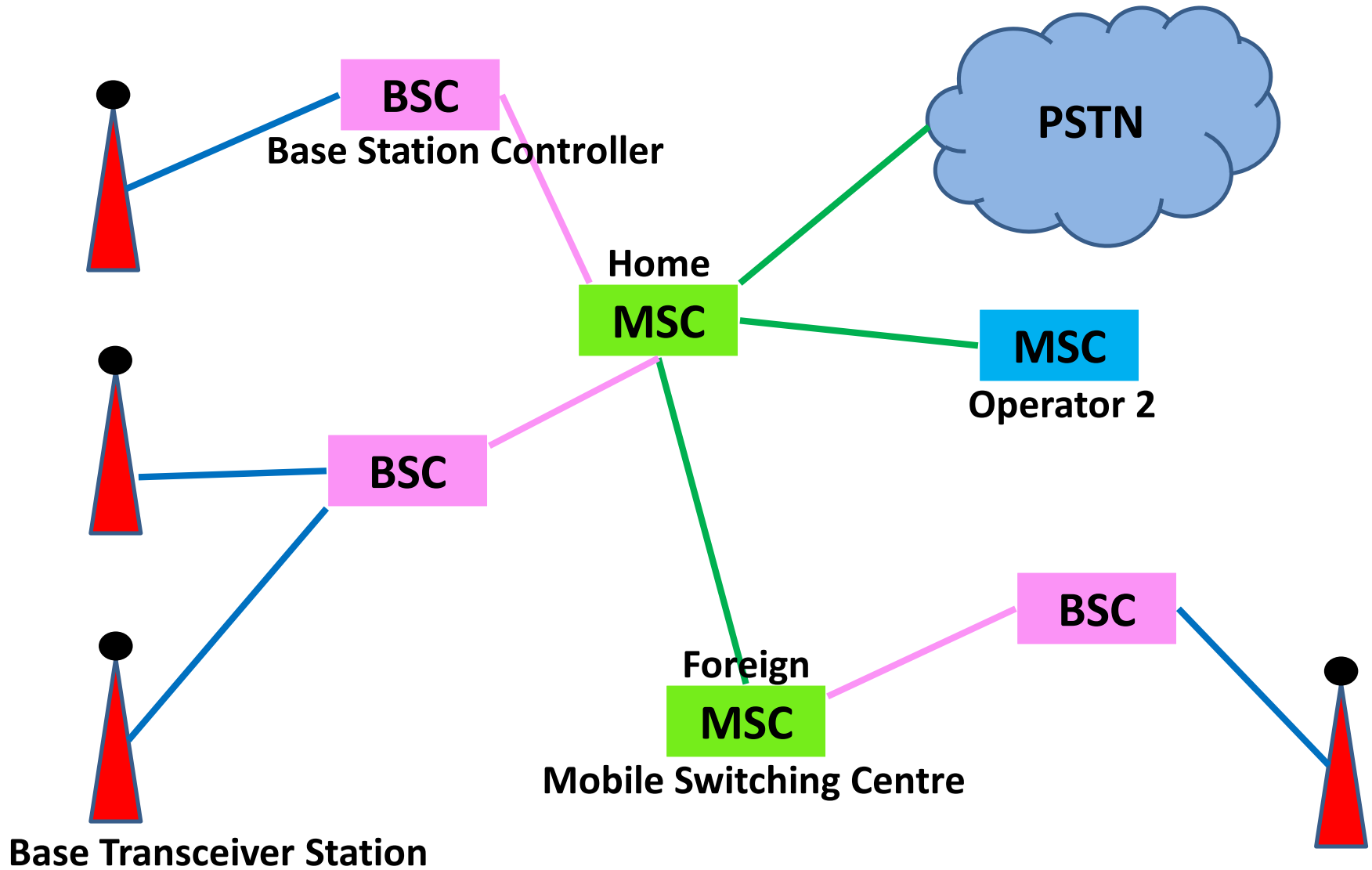
Cellular Network Architecture

Visitor Location Register (VLR)

- The VLR is a database that contains temporary information about subscribers that is needed by the MSC in order to serve visiting subscribers.
- The VLR is always integrated with the MSC. When a mobile station roams into a new MSC area, the VLR connected to that MSC will request data about the mobile station from the HLR.

How does a call take place in Cellular Network?



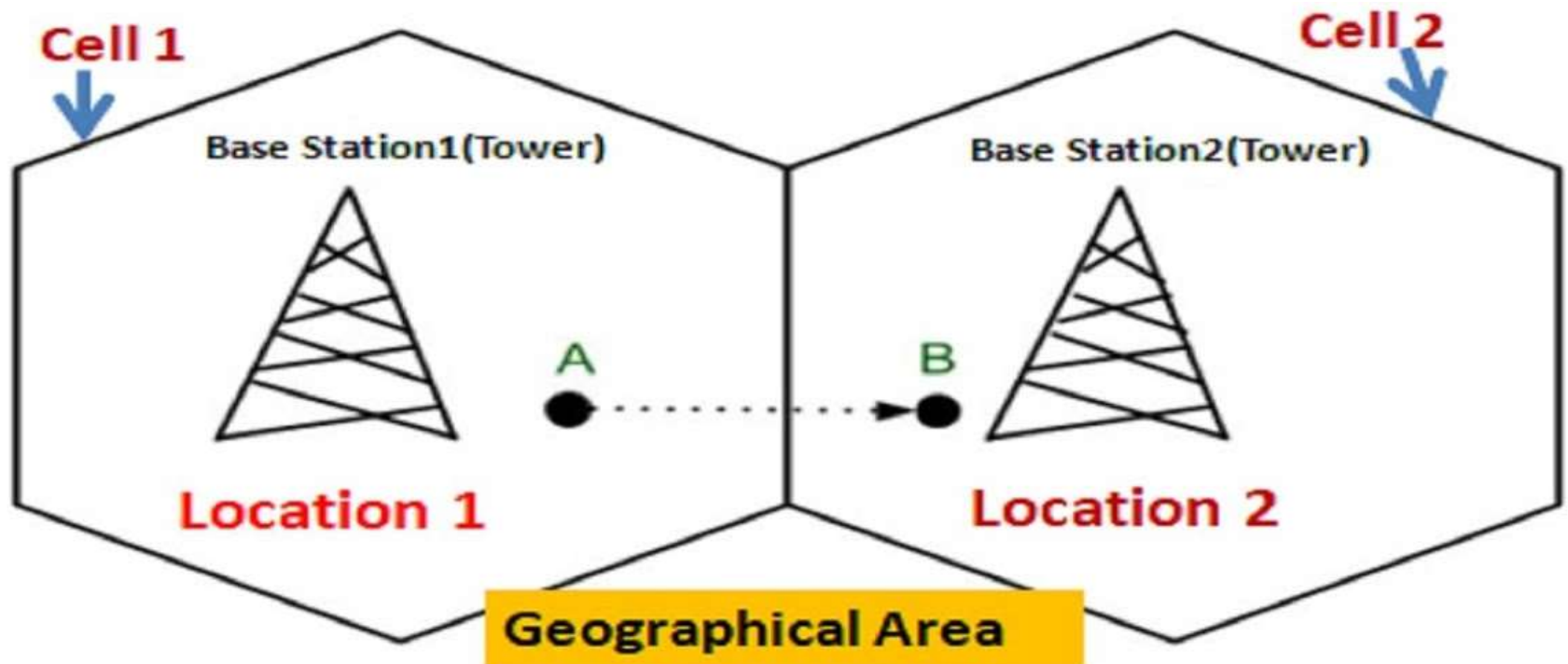


Mobility Management (Mobile Handoff)

Handoff in Cellular Network

Handoff:

- It is the ability to transfer mobile call from one Base Station to another Base Station.



Handoff in Cellular Network

Reasons of Handoff:

- Mobile device moves from one cell to another cell.
- When capacity of the cell is exhausted.
- In non CDMA, when channel used is interfered by other phone using same channel in different cell.
- In CDMA networks when there is interference from small neighbouring cells.

Handoff in Cellular Network

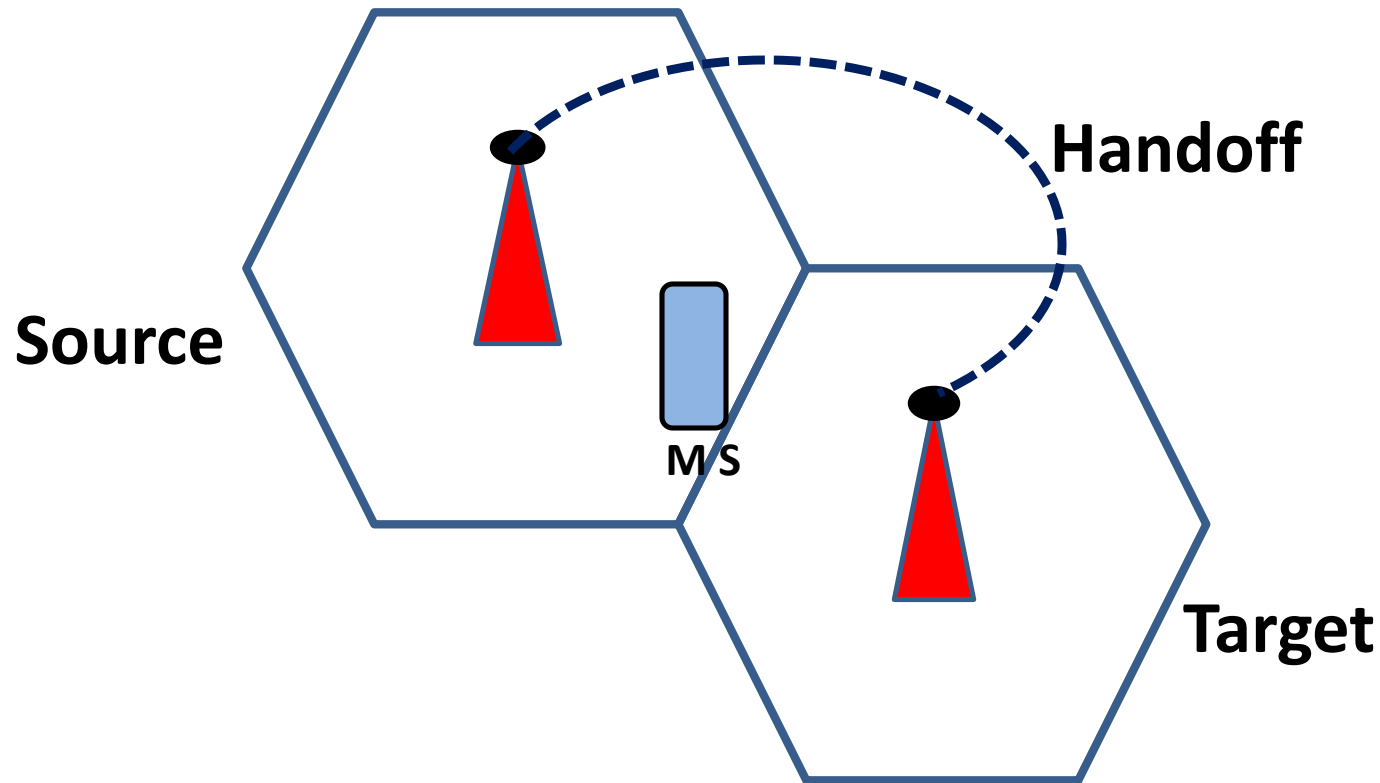
Types of Handoff:

- Inter-cell
- Intra-cell
- Hard Handoff
- Soft Handoff

Handoff in Cellular Network

Types of Handoff:

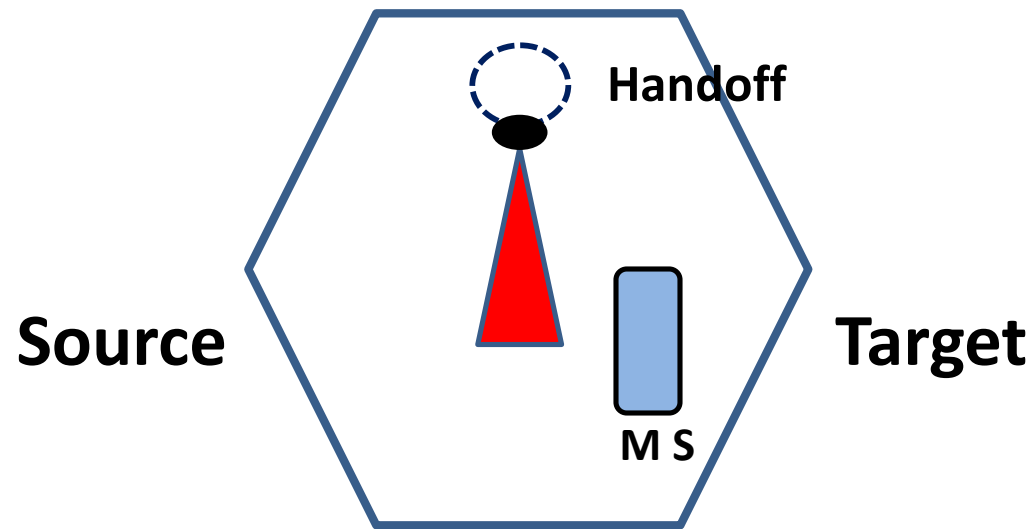
- **Inter-cell** – Source and Target cells are different.



Handoff in Cellular Network

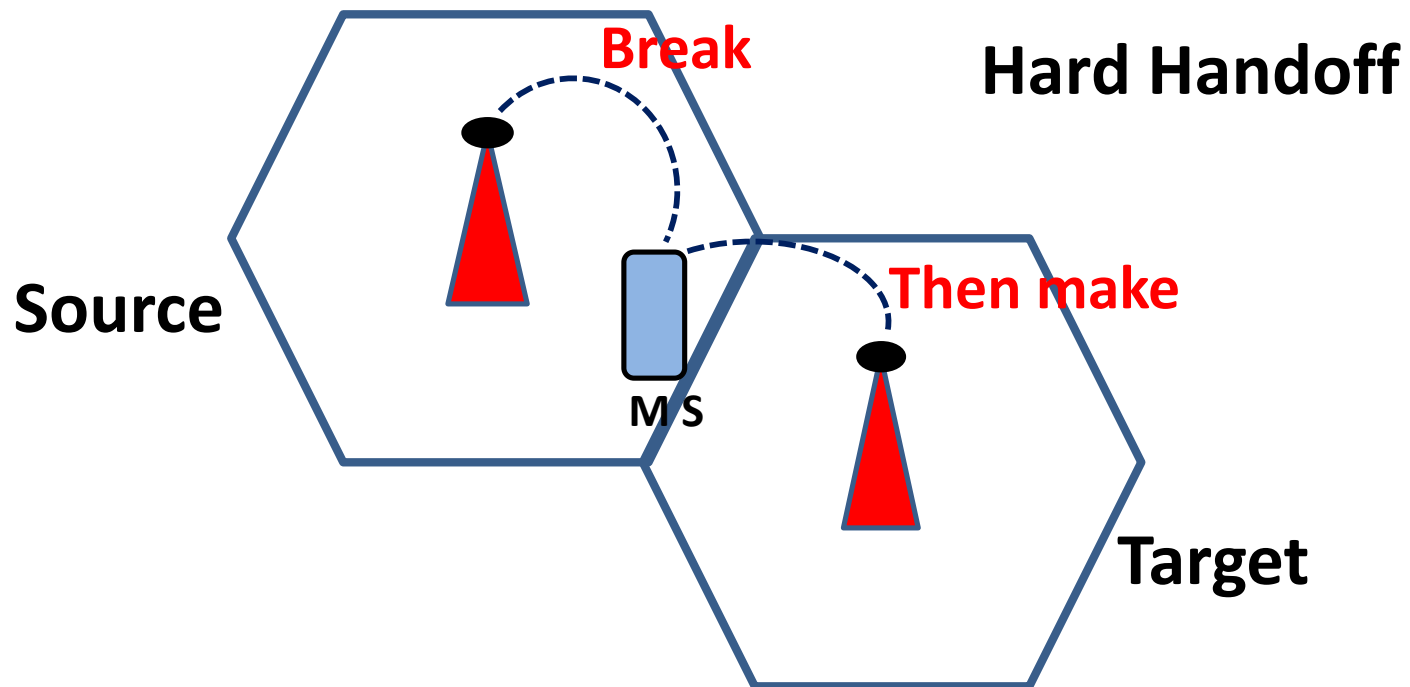
Types of Handoff:

- **Intra-cell** – Source and target cells are same but only used channel is changed due to any kind of disturbance.



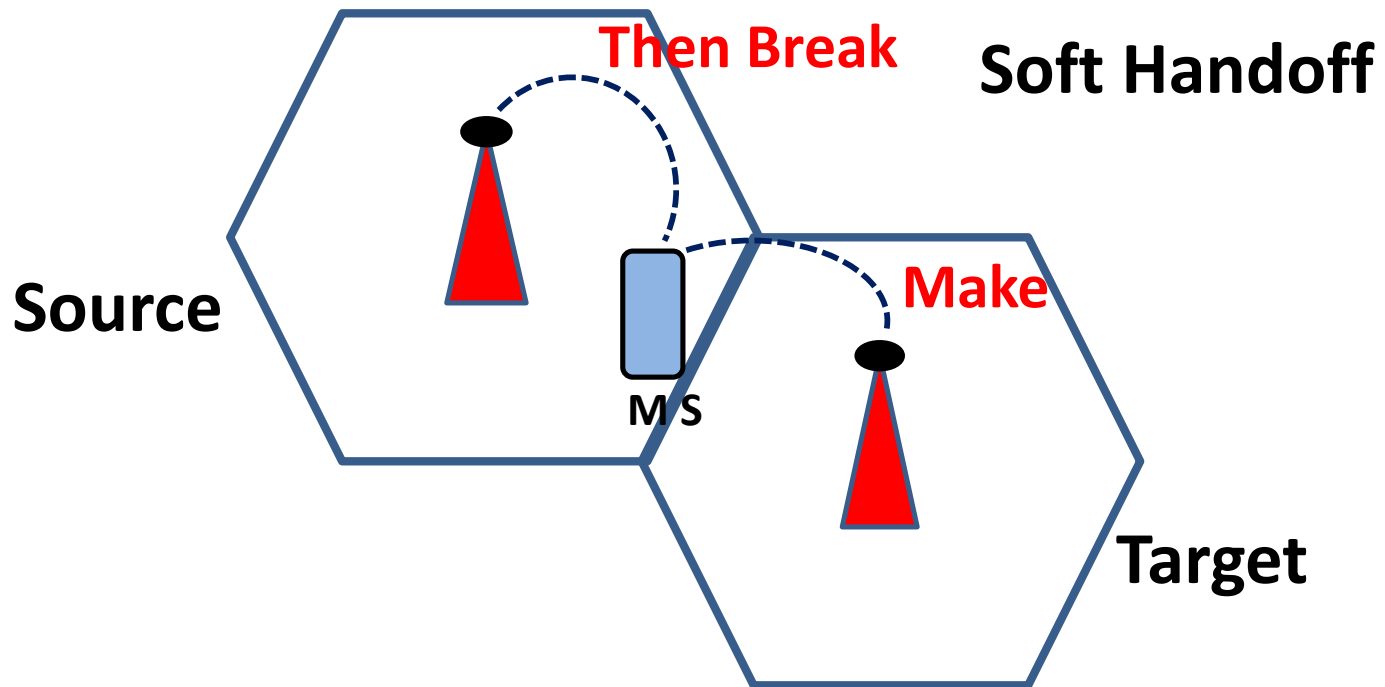
Handoff in Cellular Network

- **Hard Handoff** – Break before make.
- Channel in source cell is first released and then channel in target cell is allocated. Means only one communication at a time. It is cheap.
- **No signal for small amount of time. Disturbance is there.**



Handoff in Cellular Network

- **Soft Handoff** – Make before Break.
- Channel in source cell is retained and used for a while in parallel with channel in target cell. Two communication channels are used for some time.
- **No disturbances but expensive**



Capacity Increasing Methods

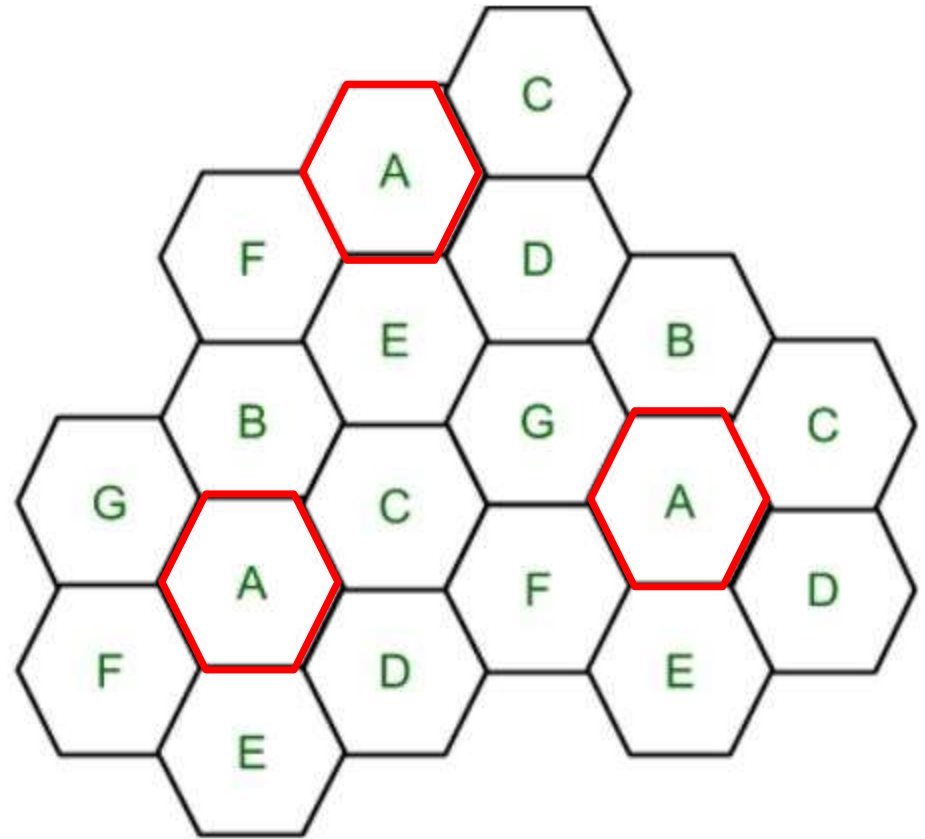
Frequency Reuse

Cell Splitting

Cell Sectoring

Frequency Reuse or Frequency Planning

- **Frequency Reuse** – It is a way to **reuse radio channels** to carry more than one conversation at a time.
- It enables to expand total system capacity without the need to employ high power antenna.
- Each cell is given few sub bands of frequency and same frequencies can be used in cells which are at distant from each other.

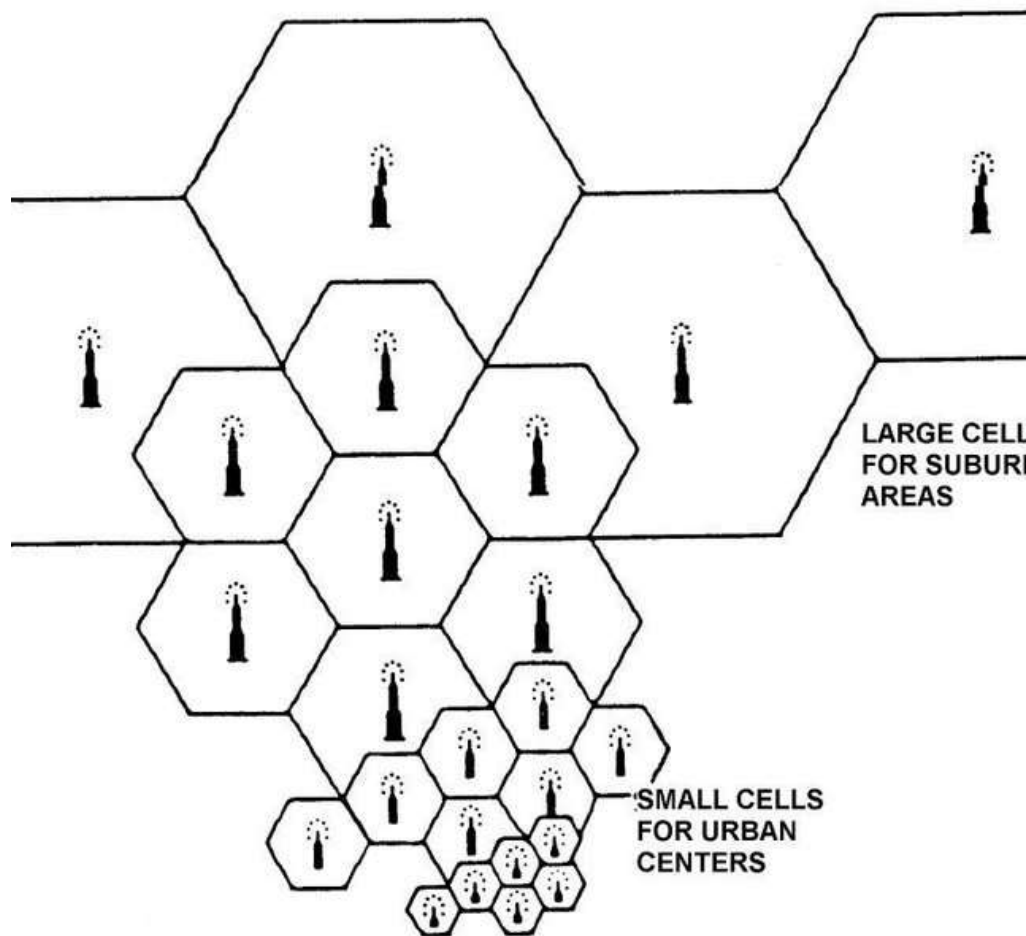


Cell Splitting

- **Cell Splitting** – It is the process of **subdividing a congested cell into smaller cells** each with its own BS.

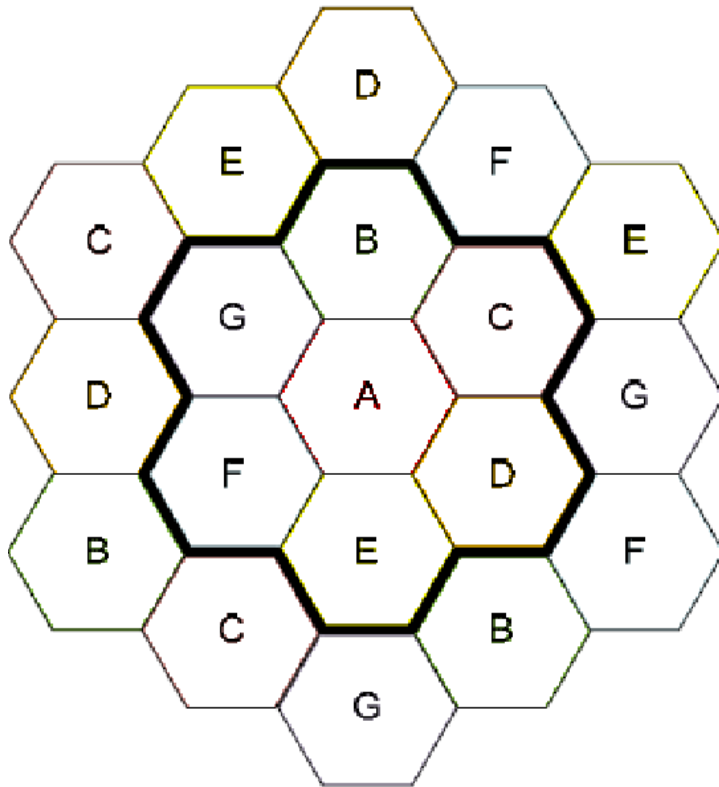
- On splitting, new cells with smaller radius are added called microcells.

- Each new cell created is independent and has reduced antenna height and transmitter power. It increases the capacity of the system as a whole.

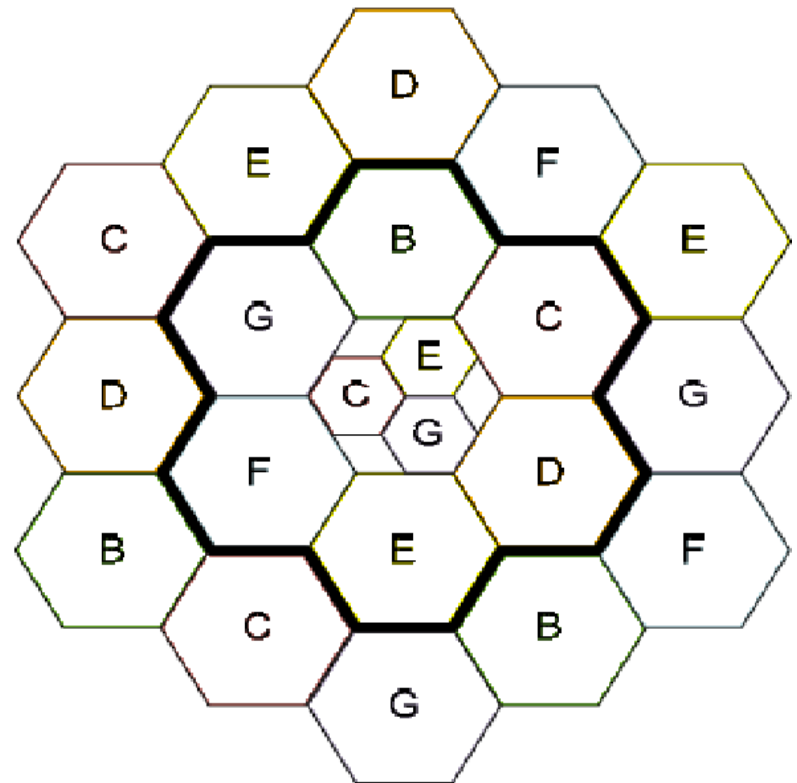


Cell Splitting

- **Advantages** – Increase the frequency reuse factor and reduces interference which drastically increases the capacity of channel.



(a) Original cell distribution



(b) After splitting cell A

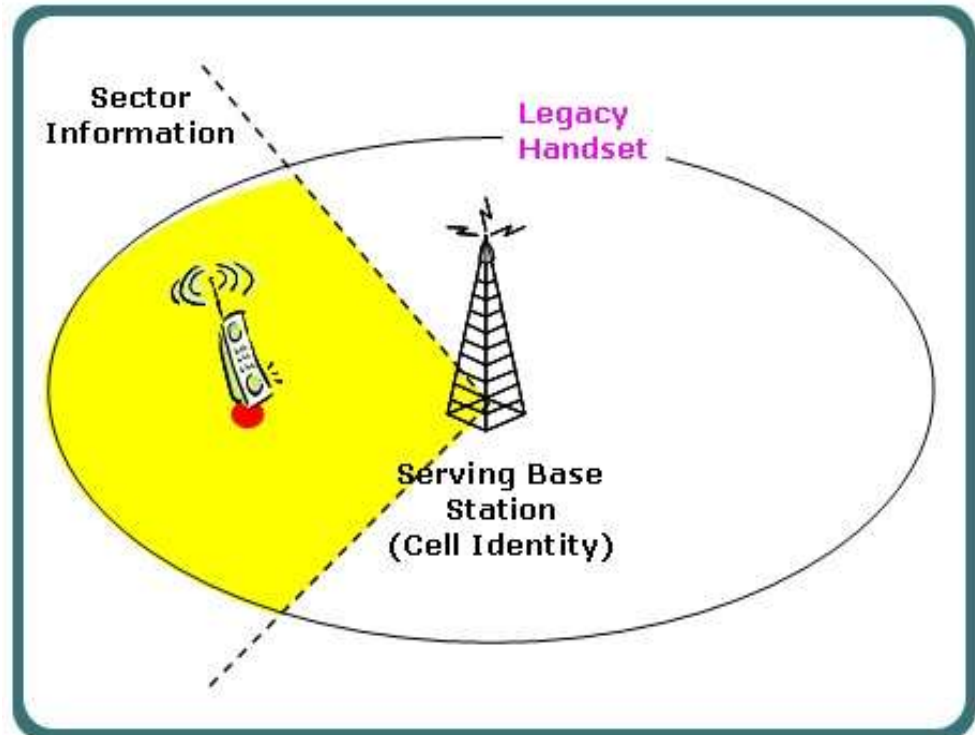
Cell Sectoring

Cell Sectoring - Cells are divided into a number of wedge-shaped sectors, each with its own set of channels. By wedge-shaped we mean that the cells are divided at an angle of 120° or 60° . These sectorized cells are called microcells.

- 3 or 6 sectors are created from a given cell.

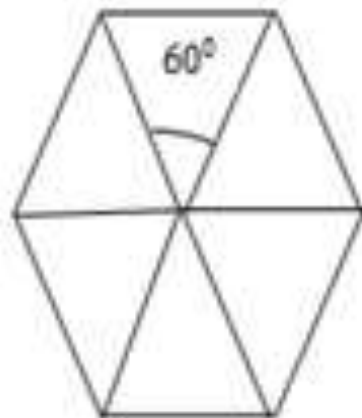
- Here the cell radius does not change after sectoring.

- It increases system performance by using a **directional antenna**.

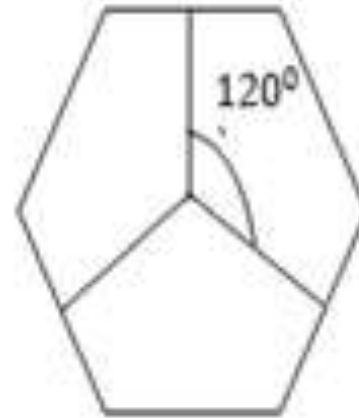


Cell Sectoring

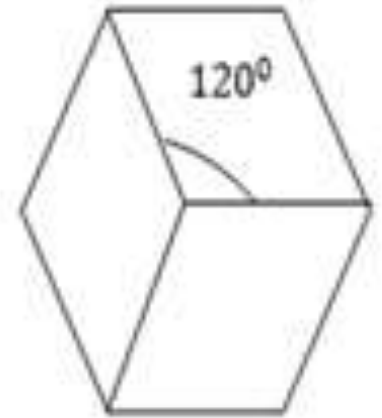
- **Advantages** – Increases frequency reuse and reduces interference which drastically increases the capacity of channel.
- **Disadvantages** - Increases the number of antennas on one BS.



60° sectoring



120° sectoring



Cell Splitting vs. Cell Sectoring

Cell Splitting	Cell Sectoring
Area of a cell is divided into micro cells	Area of a cell is divided into wedge shaped sectors
Each Micro cell has one BS and one Omni directional antenna	Each sector of a cell has same BS with one directional antenna for each sector
Radius of cell decreases	Radius of cell remains same
More processing charge	Less processing charge

Security and Interference Management (Frequency Hopping)

Frequency Hopping Spread Spectrum

•Frequency-hopping spread spectrum (FHSS)

Transmission - It is the repeated switching of the carrier frequency during radio transmission to reduce interference and avoid interception.

•**Need** – In wireless communication **interference and jamming** will slow down transmission. **Interception** is a huge security problem. To overcome these problems FHSS is used.

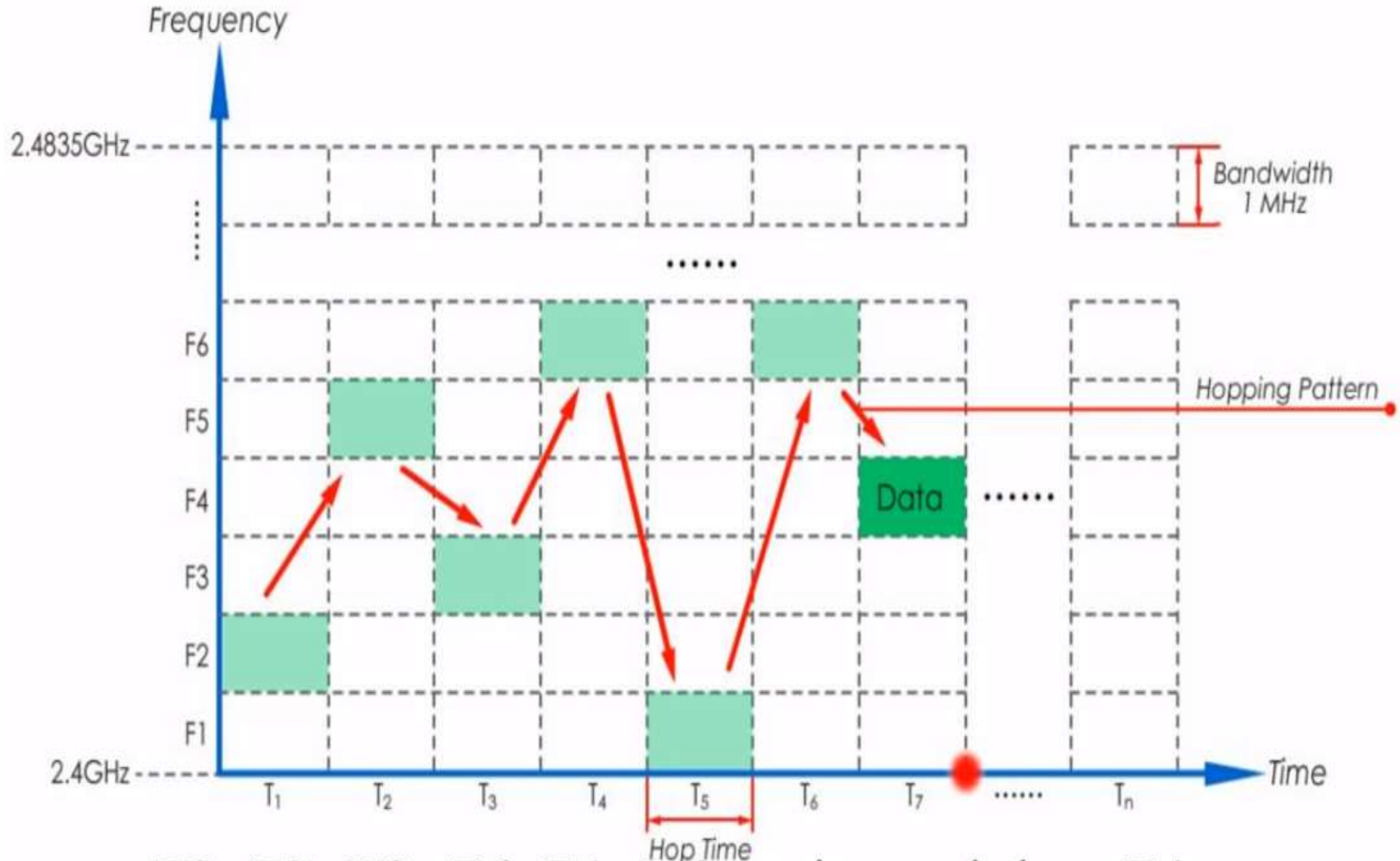
Benefits -

- reduces jamming and interference
- Prevents detection and interception

FHSS Working

- In FHSS, the transmitter hops between available narrowband frequencies within a specified broad channel in a pseudo-random sequence known to both sender and receiver.
- A short burst of data is transmitted on the current narrowband channel, and then the transmitter and receiver tune to the next frequency in the sequence for the next burst of data.
- In most systems, the transmitter will hop to a new frequency more than twice per second.

Frequency-hopping-spread-spectrum (FHSS)



F2, F5, F3, F6, F1, F6 again, and then F4.

FHSS Working

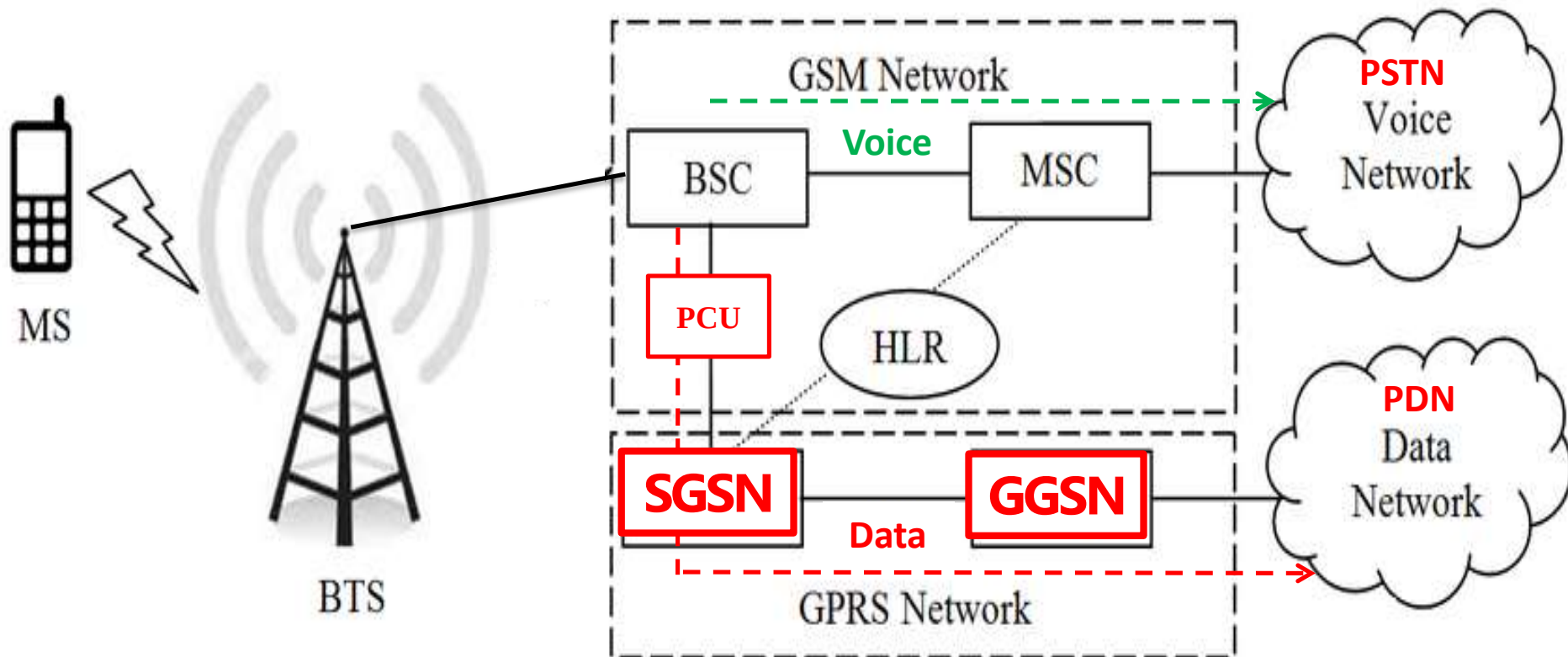
- It uses spread spectrum techniques by which a signal generated with a particular bandwidth is deliberately spread within the frequency band.
- FHSS sends data on one frequency at a time and uses different frequencies over the time.
- Hopping pattern is known only to sender and receiver.
- Different senders use different hopping pattern at different time.

GPRS

- **GPRS** stands for **General Packet Radio Service**. It is the modified version of GSM architecture service. It is non-voice, high speed packet switching technology intended for GSM networks.
- In GSM architecture we can only transport the voice signal. But if signal consists of the data then only GSM architecture cannot be used.
- For that there are two more software components used, **SGSN** (Serving GPRS supporting Node) and **GGSN** (Gateway GPRS supporting Node).

GPRS Architecture

- **GPRS** = GSM (Voice) + Data
- It is based on a modulation technique called **Gaussian minimum-shift keying (GMSK)**.



GPRS Architecture

- **PCU - Packet Control Unit.** This component is added to BSC. If signal comes to BSC and that signal contains data, then PCU routes to the SGSN.
- **SGSN - Serving GPRS Supporting Node.** It performs data compression, User Authentication and Registration. it delivers the data packet to the GGSN.
- **GGSN – Gateway GPRS Supporting Node.** GGSN routes the data packet to the data network (**PDN**).
- **PDN – Predefined Data Network** (External Data Network).

Benefits of GPRS

- It provides higher data transfer speed than fixed telecommunication networks. It is almost three times quicker than fixed-telecommunication speed.
- It provides instant connection and immediate data transfer.
- It is very cost-effective.
- It has innovative and superior applications. It provides internet applications over mobile and also facilitates Web browsing, IM messages, E-commerce etc.

GPRS is 2.5G

- Newer technologies like **EDGE** and **3G** are much faster than GPRS.
- **GPRS is considered as 2.5G** technology because it is more advanced than standard 2G digital technology, but does not meet the requirements of 3G technology.

What is needed to use GPRS ?

- An application with a GPRS modem
- A GSM/GPRS network
- A SIM card with GPRS service
- A remote station with access to the internet or the GPRS network

Satellite Communication System

**India's heaviest
communication satellite
GSAT-11 launched**



What is a Satellite?

Satellite : A satellite is a smaller object or body that revolves round a much larger object in space in a fixed well defined path. There are two type of satellites.

ARTIFICIAL SATELLITE



NATURAL SATELLITE



Types of Satellites

Natural Satellites:

- **Planets** revolving round the sun are called satellites.
- **Moon** revolving round the planet is called a satellite.

Artificial Satellites:

- They are designed and launched into space for a variety of purposes such as weather monitoring, navigation, tv and mobile communication, planetary research etc. Ex: **INSAT, IRS, GSAT**, etc.

Basic Elements of Satellite Communication System

**EARTH
STATION**

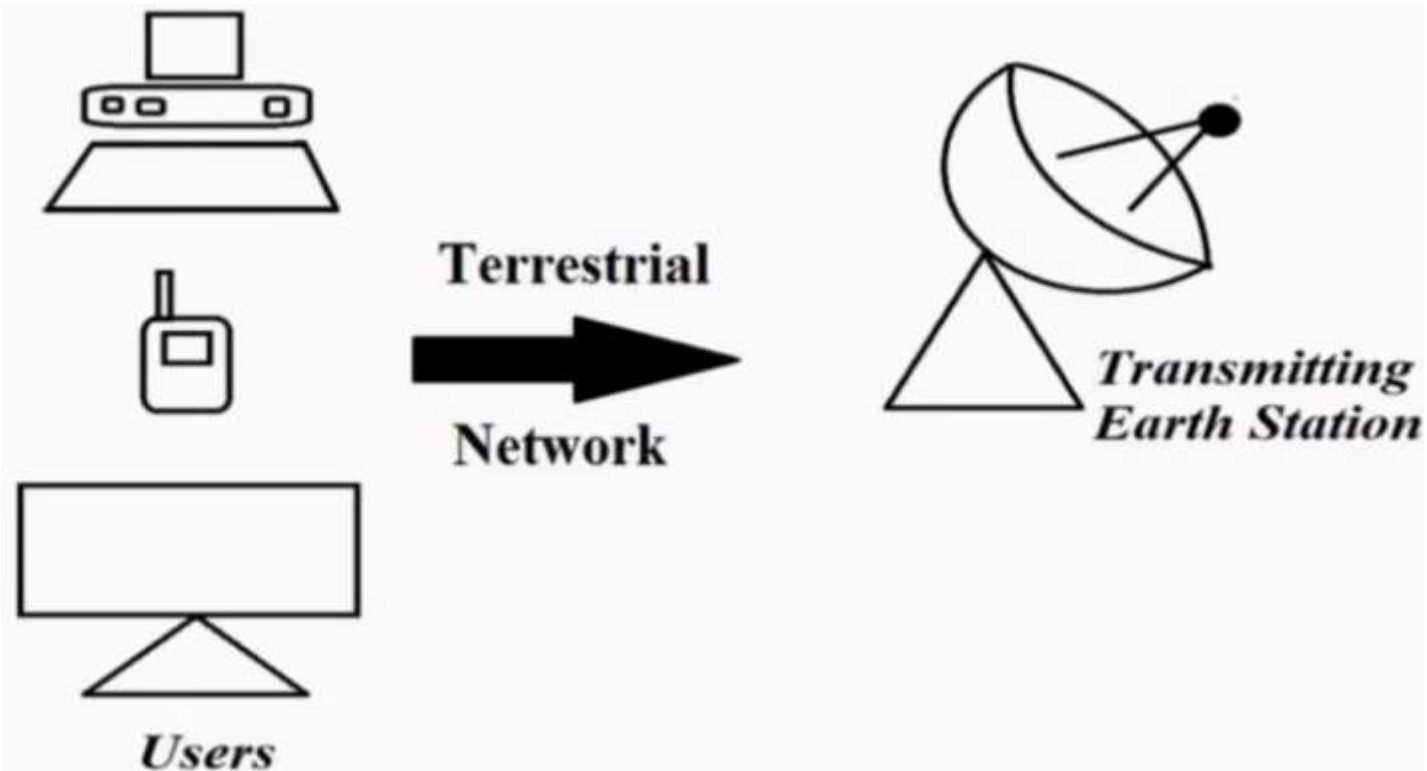
SATELLITE

**TERRESTRIAL
SYSTEM**

USER

Working Principle of a Satellite

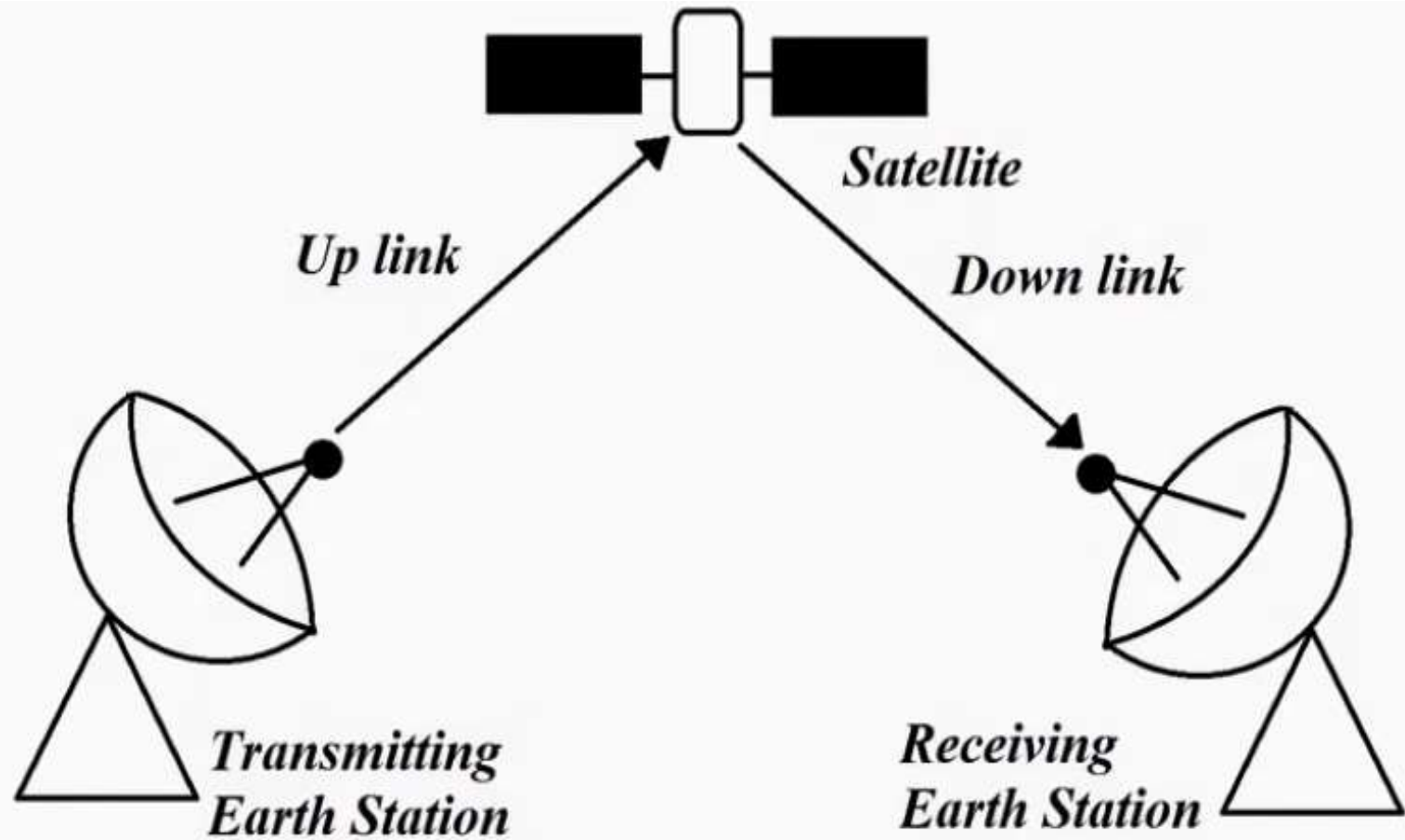
- The network contains several Earth Stations on Earth which are linked with a Satellite in Space.
- The End users are connected to these Earth Station through a Terrestrial Communication Network.



Working Principle of a Satellite

- The user generates a message signal which is passed and processed through the Terrestrial Communication Network to the Earth's Station.
- The Earth Station modulates the signal and sends that signal to the satellite in space on the **uplink frequency spectrum**.
- The satellite amplifies and processes the signal and retransmits it back on the **downlink frequency spectrum**.

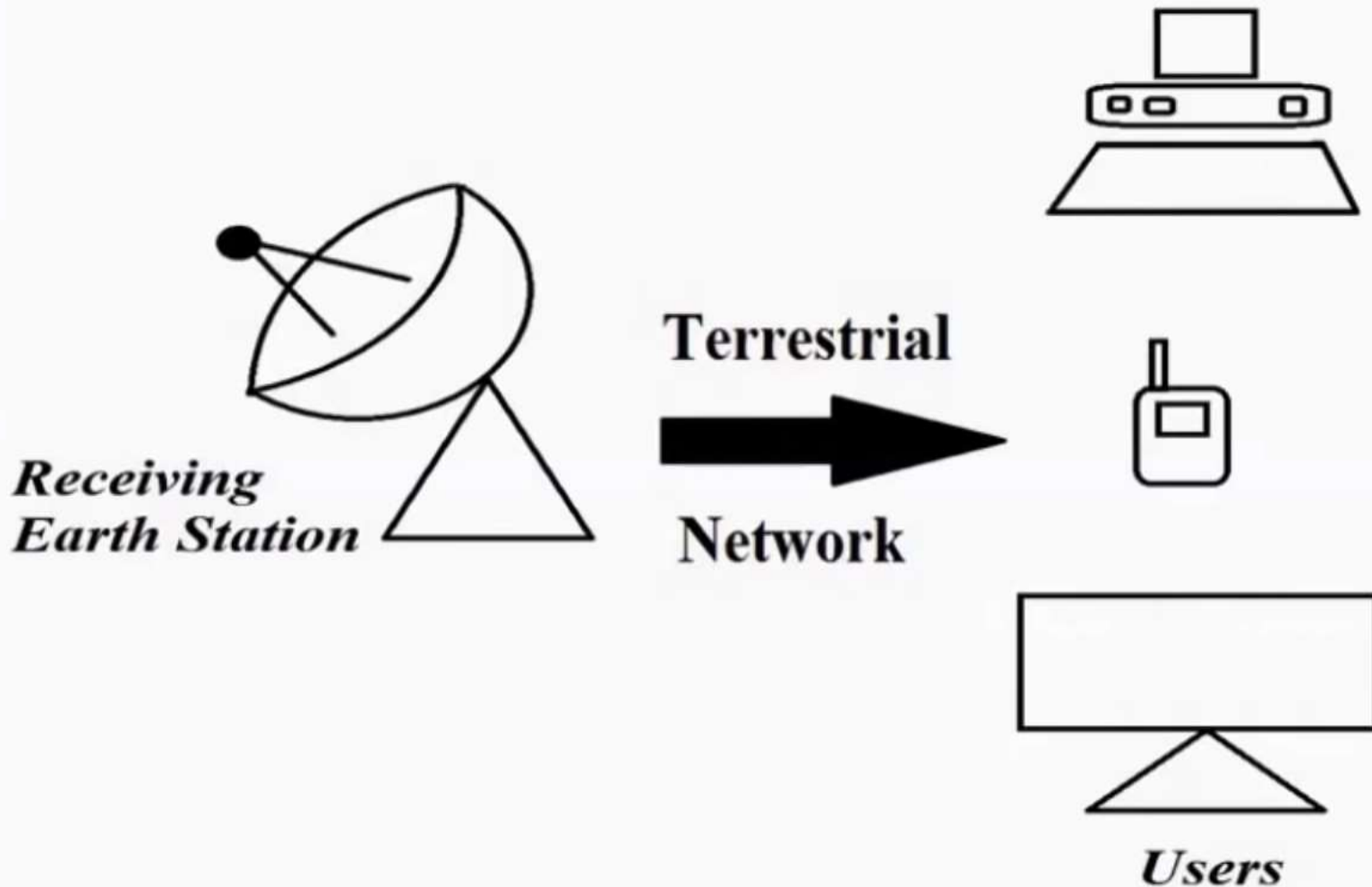
Working Principle of a Satellite



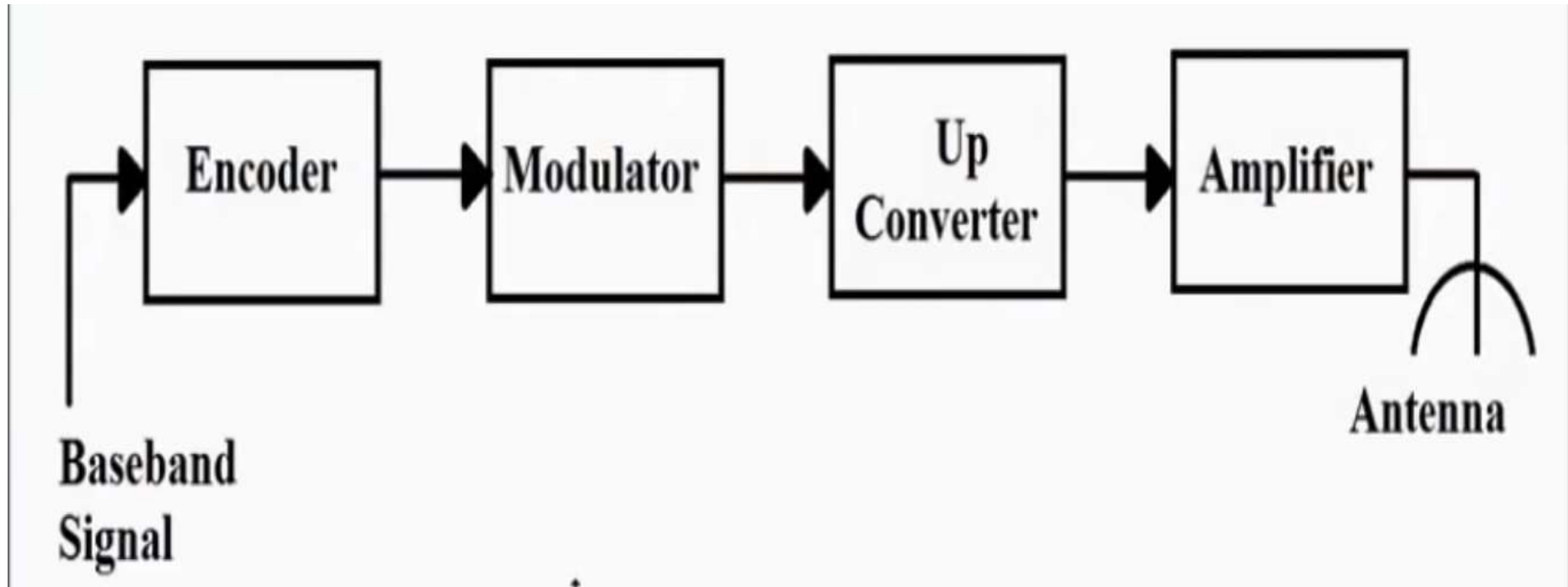
Working Principle of a Satellite

- At the receiving Earth Station the retransmitted signal is received , demodulated and amplified to get back the original signal.
- The signal is then send to the user through the Terrestrial Communication Network.

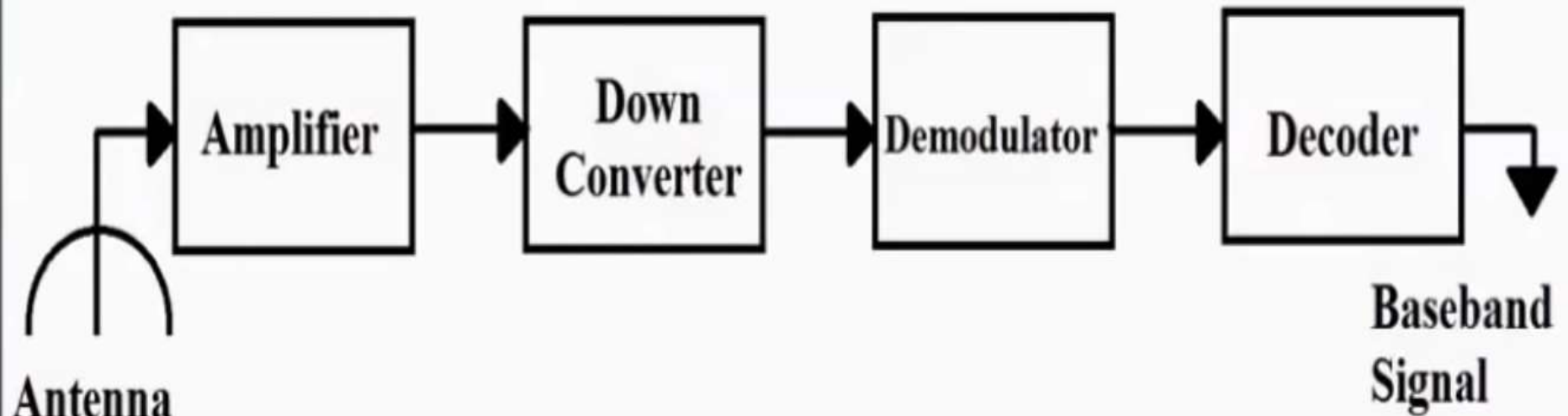
Working Principle of a Satellite



Block Diagram of Transmitter in Satellite



Block Diagram of Receiver in Satellite



Benefits of Satellite Communication

- **Broad coverage:** Satellites can provide communication across large areas, including remote regions
- **Reliability:** Satellites can provide reliable communication even when cellular networks are unavailable
- **Speed:** Satellites can provide high-speed broadband connections, which are useful for video streaming and large file transfers
- **Flexibility:** Satellites can be used for a variety of purposes, including broadcasting, navigation, and

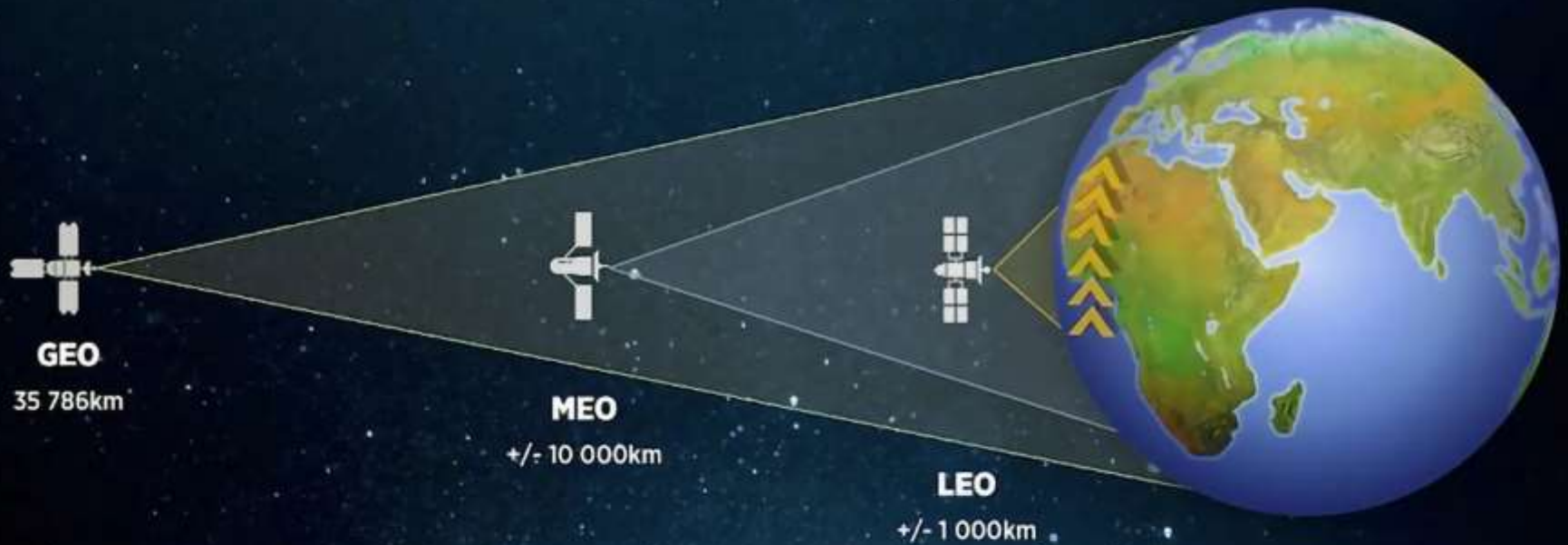
Benefits of Satellite Communication

- Environmental monitoring:** Satellites can provide information on a wide area at once, which can help with weather forecasting, disaster monitoring etc.
- Economic support:** Satellite data can help support the economy
- Public safety:** Satellite data can help ensure public safety and national defense
- Emergency management:** Satellite phones can provide communication in areas where cellular networks are unreliable

Satellite Orbits



Satellite Orbits



Satellite Orbits

Satellites can orbit the Earth in three main ways:

- **Geostationary Earth Orbit (GEO):** Satellites in this orbit appear fixed in the sky to an observer on Earth.
- **Medium Earth Orbit (MEO):** Satellites in this orbit are between low Earth orbit and geostationary orbit.
- **Low Earth Orbit (LEO):** Satellites in this orbit are closest to Earth.

Working of Satellite Orbits

- **GEO** - Satellites in this orbit revolve around the Earth once per day over the equator.
- **MEO** - Satellites in this orbit have orbital periods ranging from about 2 to 14 hours.
- **LEO** - Satellites in this orbit have a faster revolving period because they are closer to Earth.

Benefits of Satellite Orbits

LEO (Low Earth Orbit):

- **Low Latency:** Due to their less distance from Earth, LEO satellites offer minimal signal travel time, making them excellent for real-time applications like video conferencing and online gaming.
- **High Bandwidth Potential:** With a constellation of LEO satellites, large amounts of data can be transmitted quickly.
- **Improved Coverage in Remote Areas:** Can provide connectivity to regions with limited terrestrial infrastructure.

Benefits of Satellite Orbits

MEO (Medium Earth Orbit):

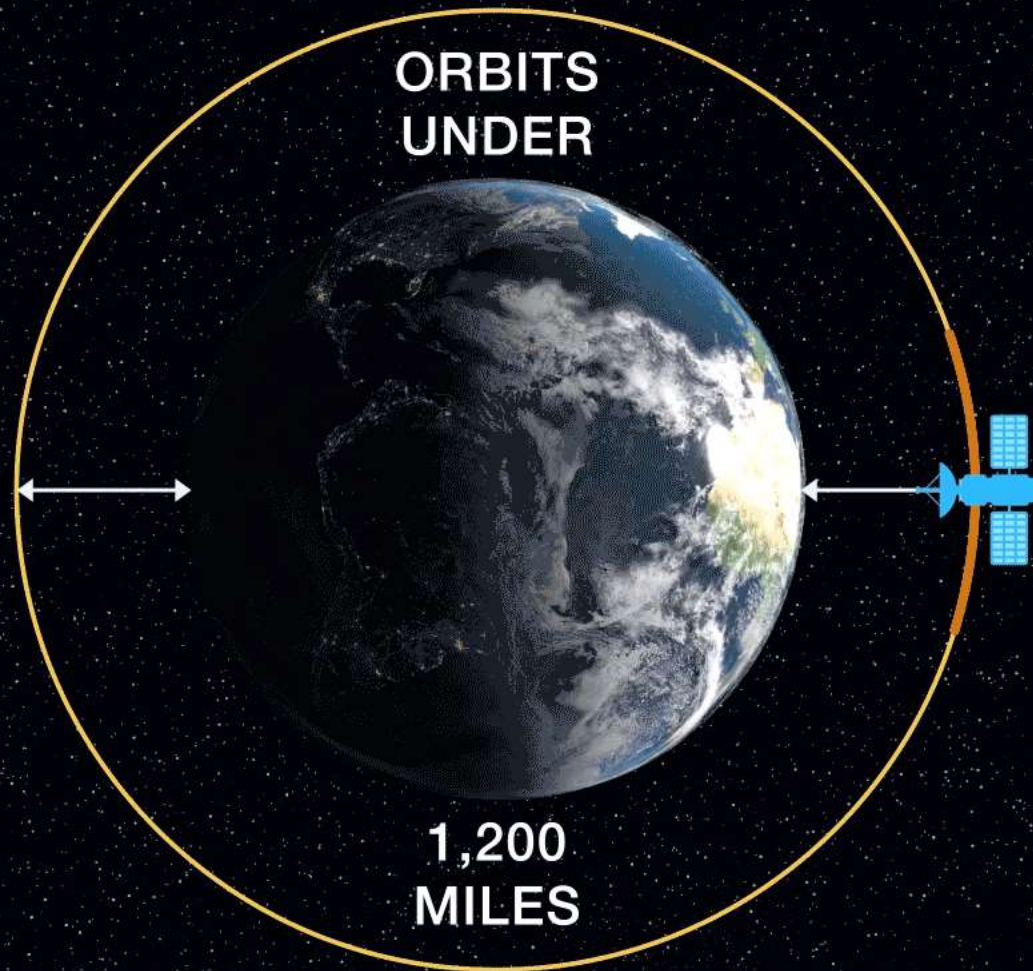
- **Balance of Latency and Coverage:** Offers a compromise between the low latency of LEO and wide coverage of GEO, making them suitable for applications like maritime communication and navigation.
- **Flexibility in Constellation Design:** Can be configured to provide targeted coverage for specific regions.

Benefits of Satellite Orbits

GEO (Geostationary Earth Orbit):

- **Continuous Coverage over a Fixed Location:** Since GEO satellites stay positioned over a specific point on Earth, they provide uninterrupted coverage for broadcasting and weather monitoring.
- **Fewer Satellites Needed for Global Coverage:** A small constellation of GEO satellites can provide near-global coverage.
- **High Power Transmission:** Capable of delivering strong signals over large distances.

1 ORBIT EVERY 90 MINUTES



LOW EARTH ORBIT (LEO)

Starlink Network

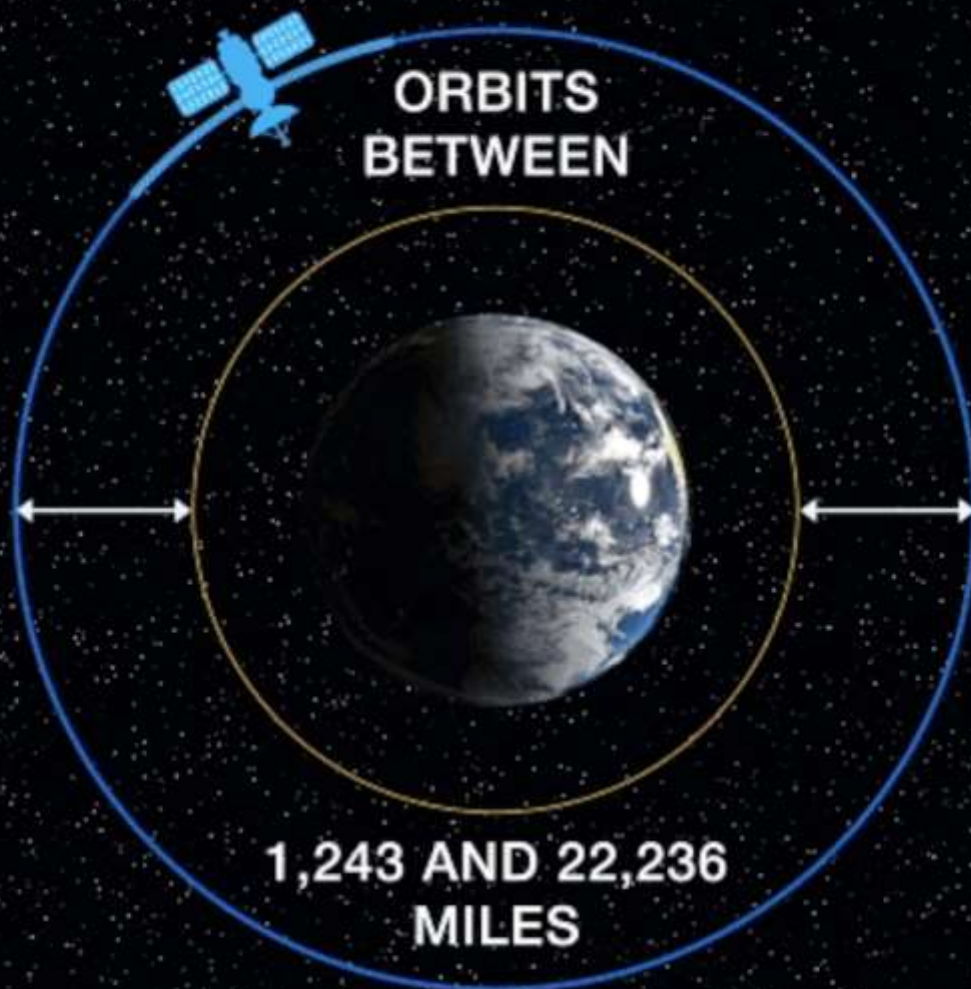


Uses of LEO

LEO – Communication, Earth observation, scientific research and spying.

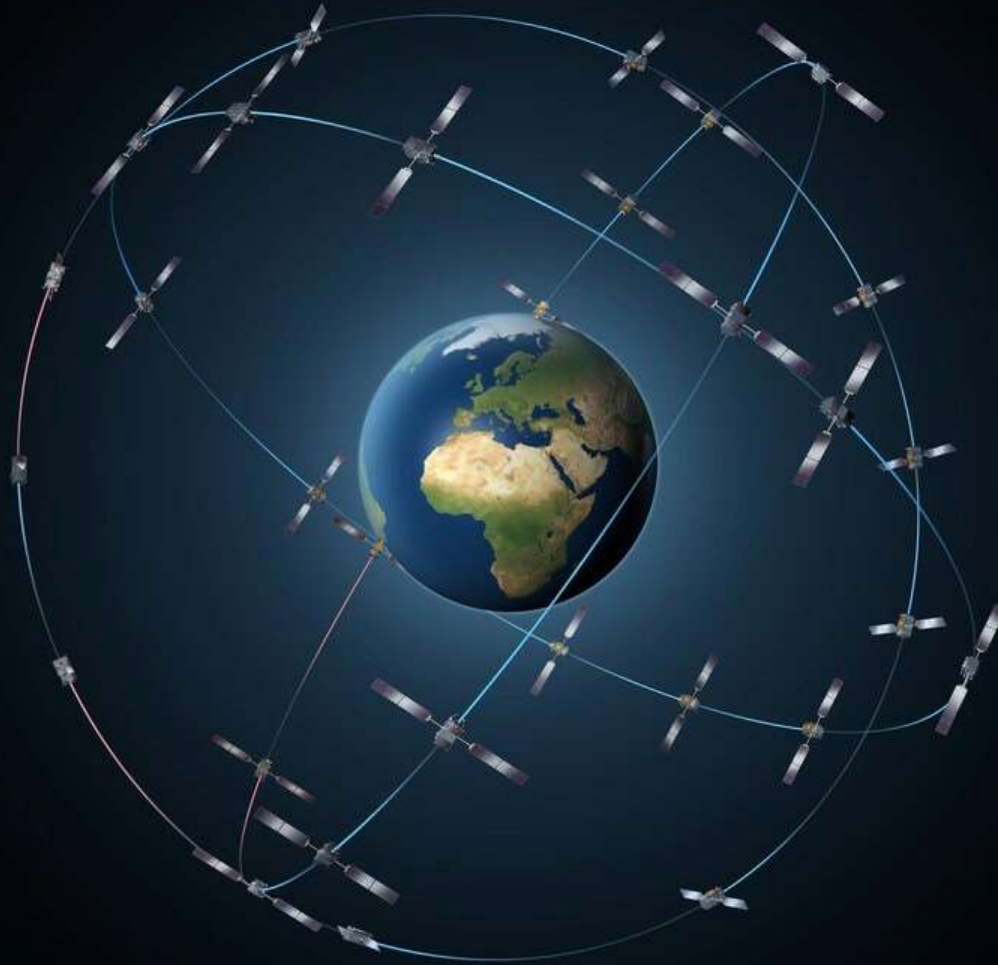
- **Altitude range:** Orbits the Earth at an altitude of 2,000 kilometers (1,200 miles) or less.
- **Popular satellites using LEO:** International Space Station (ISS), the Hubble Telescope, and Starlink satellites.

1 ORBIT EVERY 12 HOURS



MEDIUM EARTH ORBIT (MEO)

Satellite Network in MEO

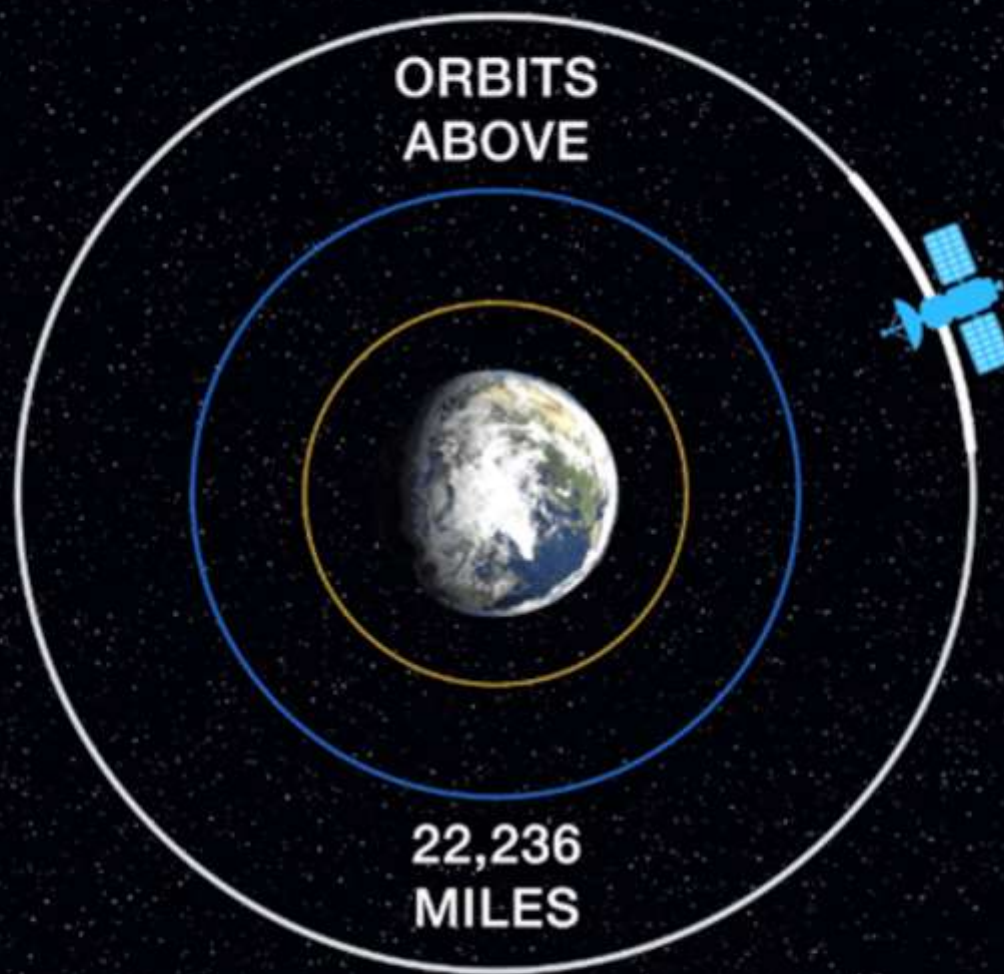


Uses of MEO

MEO – Navigation, Rural area Communication, Remote Sensing

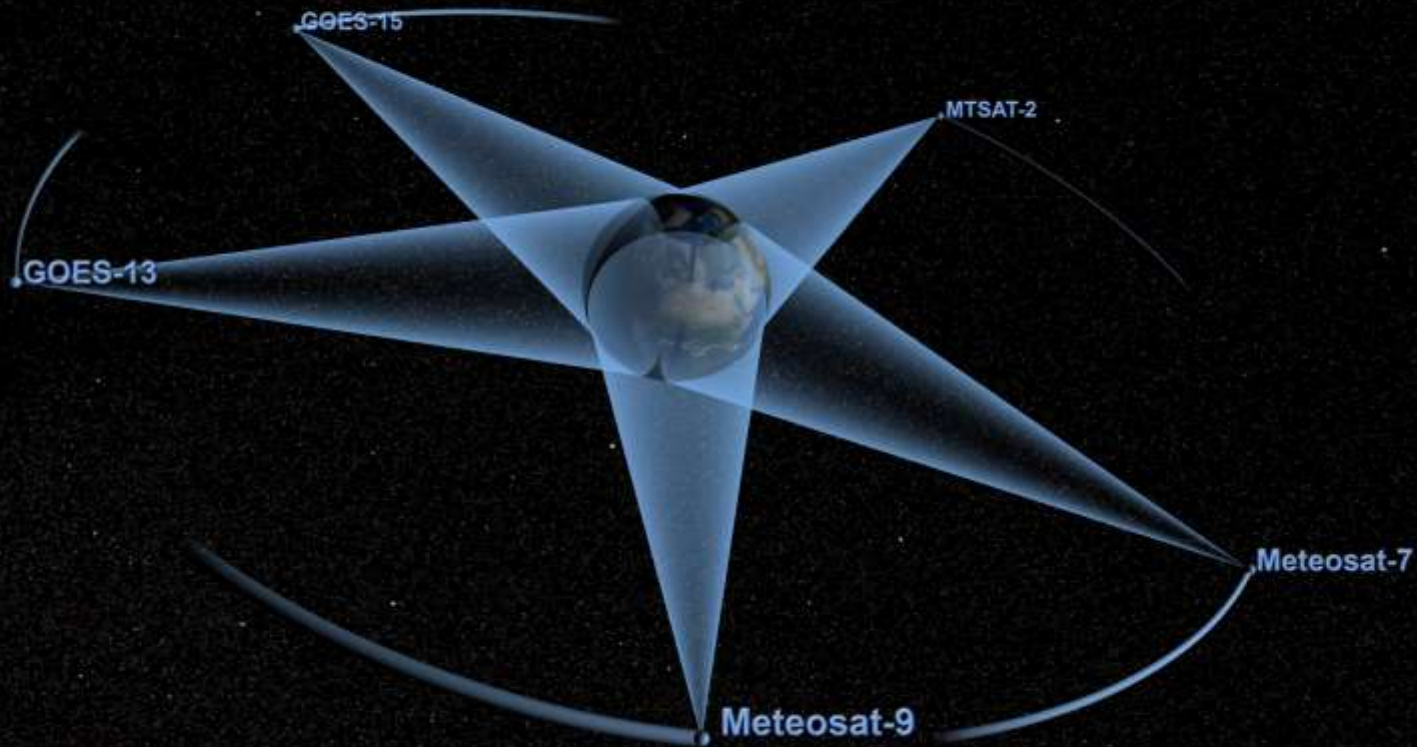
- **Altitude range:** Typically between 2,000km and 35,786km (1,243 miles and 22,236 miles) above Earth's surface.
- **Popular navigation systems using MEO:** GPS (Global Positioning System), Galileo (European Union), GLONASS (Russia).

1 ORBIT EVERY 24 HOURS



**GEOSYNCHRONOUS
EARTH ORBIT (GEO)**

Satellite Network in GEO



Uses of GEO

GEO – Telecommunications, weather monitoring, remote sensing, defense and security

Altitude range: Height of 36,000 km (above 22,236 miles) from the equator of Earth.

Popular GEO satellites – INSAT (Indian National Satellite System for telecom industry), GSAT (Two way communication for distant education), GSAT-7 (For Indian defense system)

GEO Synchronous means GEO Stationary

Satellite Uplink Downlink Frequency Range

C-band, Ku-band, and Ka-band are all frequency bands used for satellite communications. Each have different uses depending on the application, location, and regulatory requirements.

Frequency Band	Downlink Range	Uplink Range
C	3.7 GHz – 4.2 GHz	5.925 GHz – 6.425 GHz
Ku	11.7 GHz – 12.2 GHz	14.0 GHz – 14.5 GHz
Ka	17.7 GHz – 21.2 GHz	27.5 GHz – 31.0 GHz

Satellite Uplink Downlink Frequency Range

C-band – Used for mobile communications, and weather radars.

Ku-band – satellite TV, broadband internet, and corporate networks.

Ka-band – High speed broadband internet, Police Radar for a narrower target focus, Radar for high resolution images

Local Loop

- **Local loop** is a circuit line from a subscriber's phone to the local central office (LCO).
- **The implementation of local loop of wires is costly** for the operators, especially in rural and remote areas due to less number of users and increased cost of installation.
- Hence, **the solution for it is the usage of wireless local loop (WLL)** which uses wireless links rather than copper wires to connect subscribers to the local central office.

Wireless Local Loop

- **WLL** : Wireless Local Loop is a two-way communication service to a user near small service area.
- It uses Radio Signals and Wireless Networking Technologies to local loop is used instead of copper cable technology for access to public switched telephone network (PSTN)

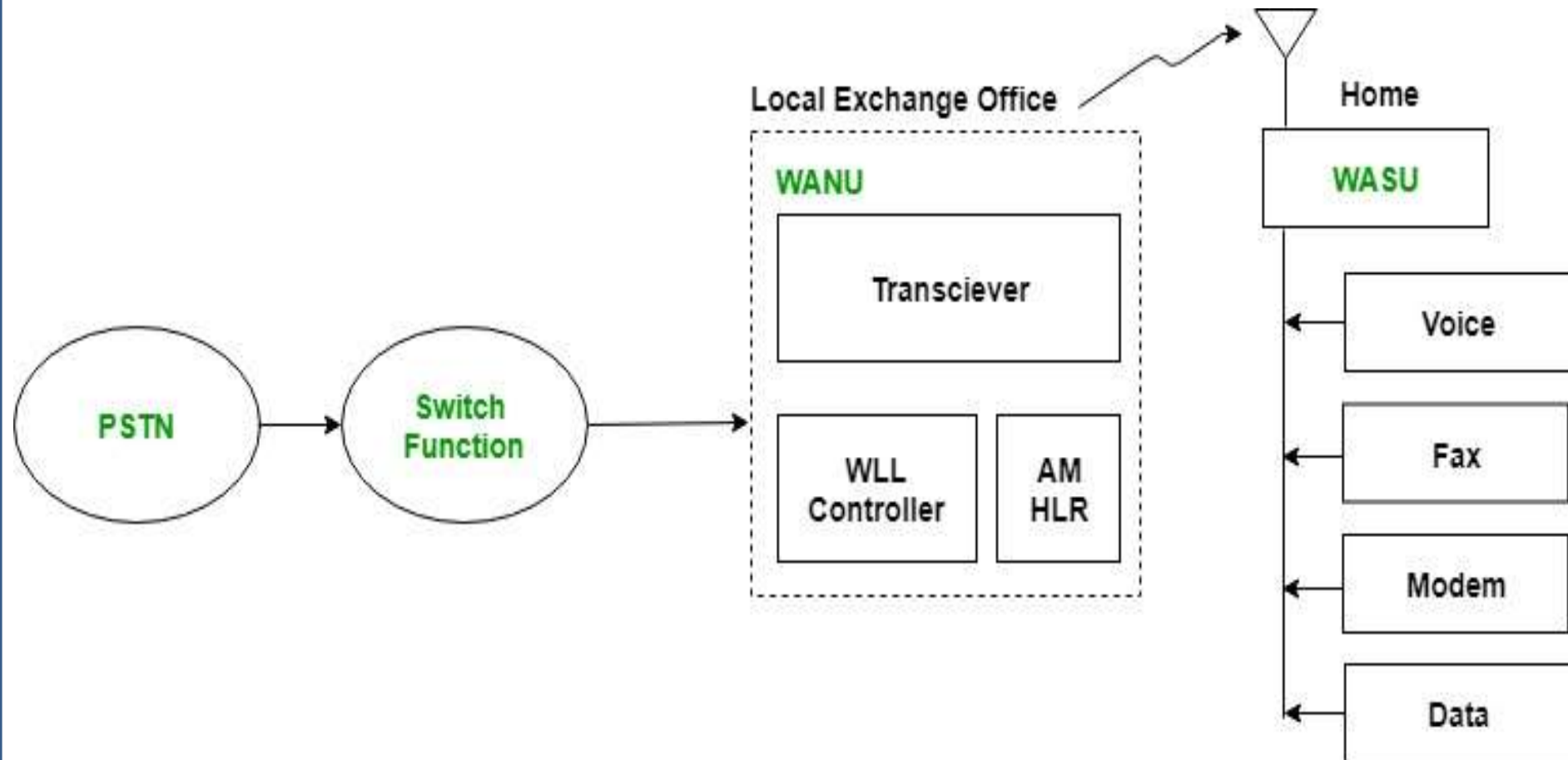
Advantages of WLL

- It eliminates the first mile or last mile construction of the network connection.
- **Low cost** due to no use of conventional copper wires.
- **Much more secure** due to digital encryption techniques used in wireless communication.
- **Highly scalable** as it doesn't require the installation of more wires for scaling it.

Features of WLL

- **Audio Service**
- **Data Service**
- **Fax Service**
- **Internet connection via modem**

WLL Architecture



WLL Components

- **PSTN** - It is Public Switched Telephone Network which is a circuit switched network. It is a collection of world's interconnected circuit switched telephone networks.
- **Switch Function** - Switch Function switches the PSTN among various WANUs.
- **WASU** - Wireless Access Subscriber Unit is present at the house of the subscriber. It connects the subscriber to WANU and the power supply for it is provided locally.

WLL Components

WANU - Wireless Access Network Unit is present at the local exchange office. All local WASUs are connected to it. Its functions includes: Authentication, Operation & maintenance, Routing, Transmitting voice and data. It consists of following sub-components:

Transceiver: It transmits/receives data.

WLL Controller: It controls the wireless local loop component with WASU.

AM: Access Manager is responsible for authentication.

HLR: Home Location Register stores the details of all local WASUs.